Faculty of Engineering

Topics in Computer Science
Constraint Programming

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SCHEDULE

Scheduling



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Recent Publications

- Complexity of n-Queens completion (extended abstract)
- A framework for constraint based local search using ESSENCE
- Minimal and canonical images



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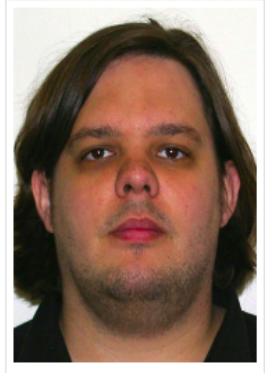
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1 Introduction
1.1 Another Example: Graph Coloring
1.2 Variables, Domains and Constraints
1.3 Solving:
Brute Force
Backtracking

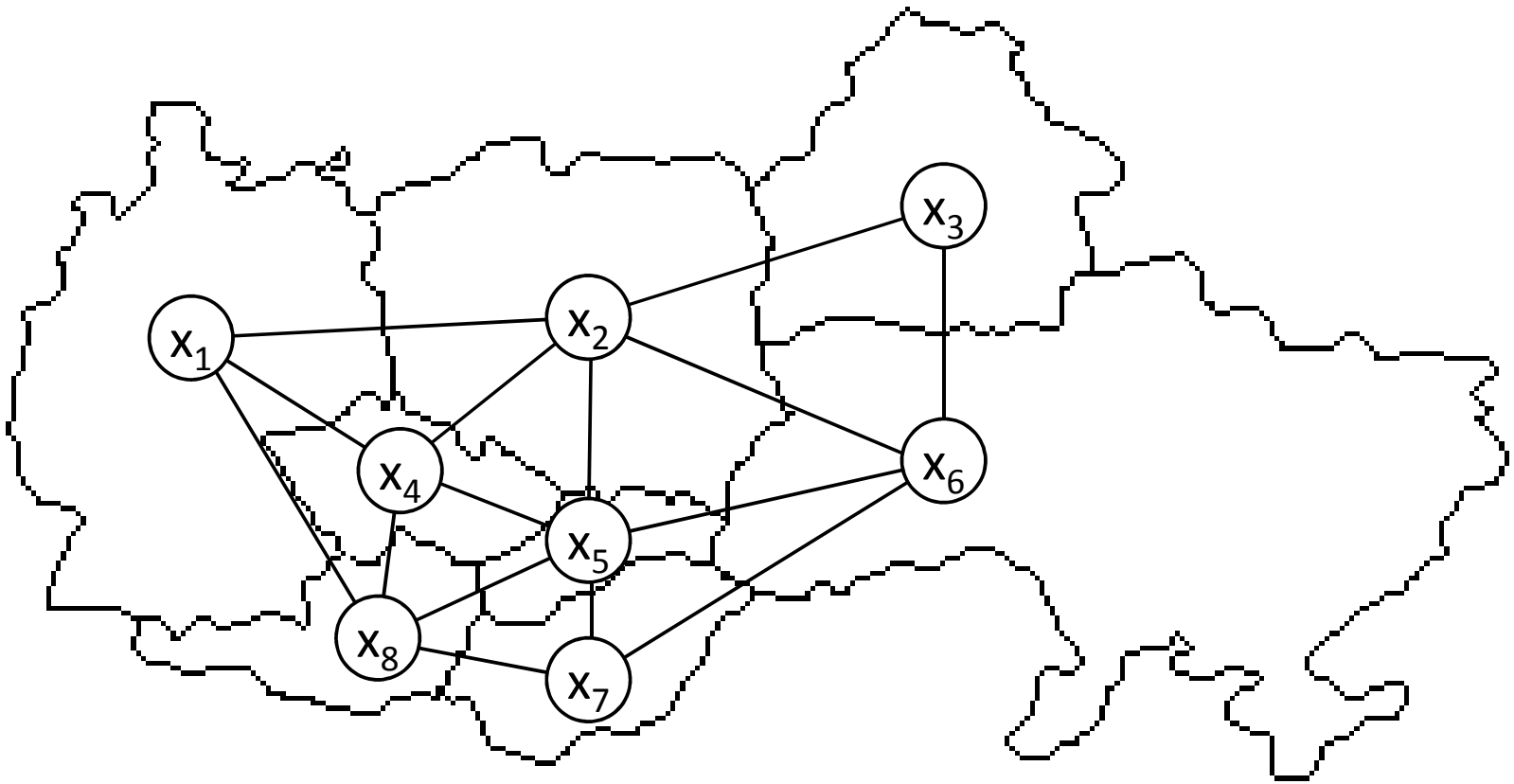
2 Constraint Programming
2.1 CSP
2.2 CP solving
2.3 History
2.4 Solvers

3 Examples

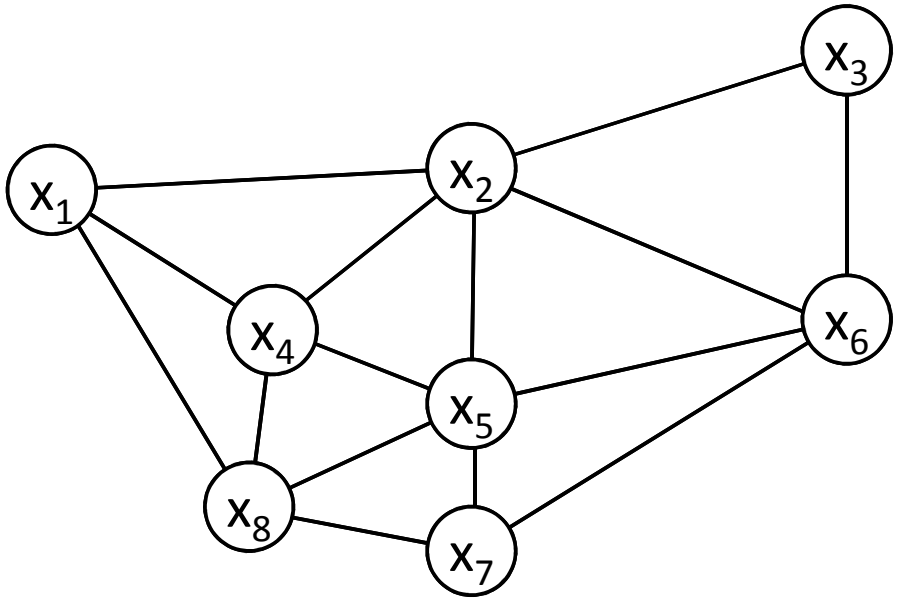
4 Applications

5 Conclusion

Introduction



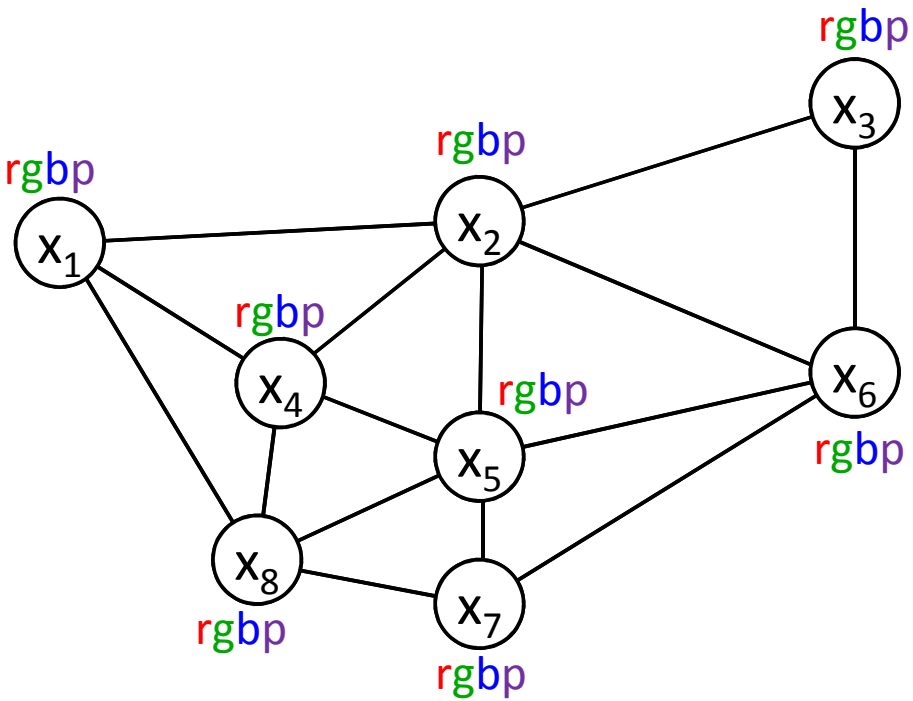
27



Variables and domains: x_i in {r,g,b,p} for all i

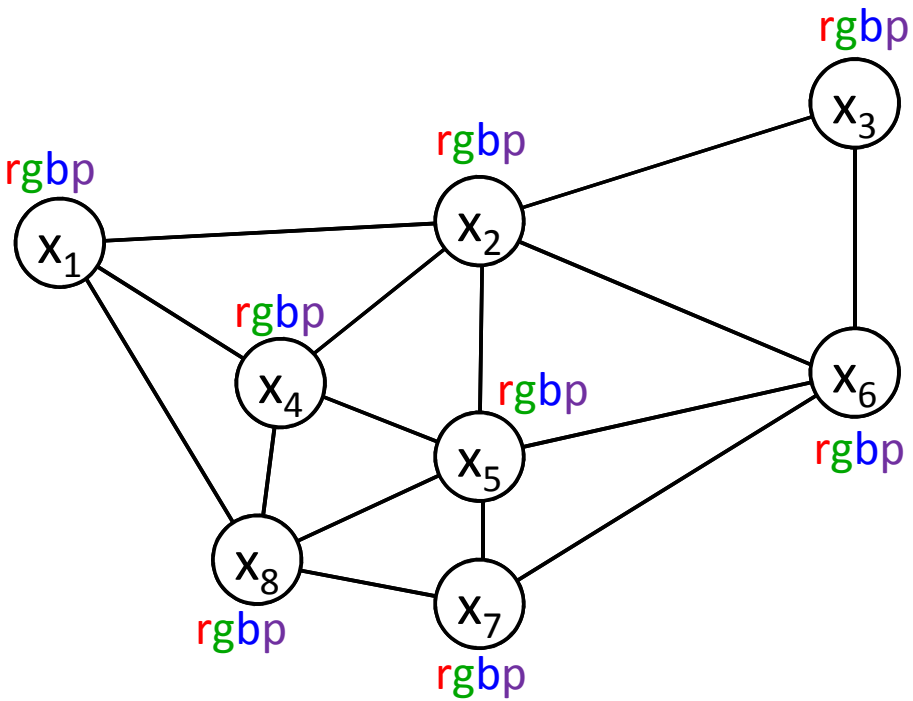
Constraints: $x_i \neq x_j$ for all edges (i,j)

28



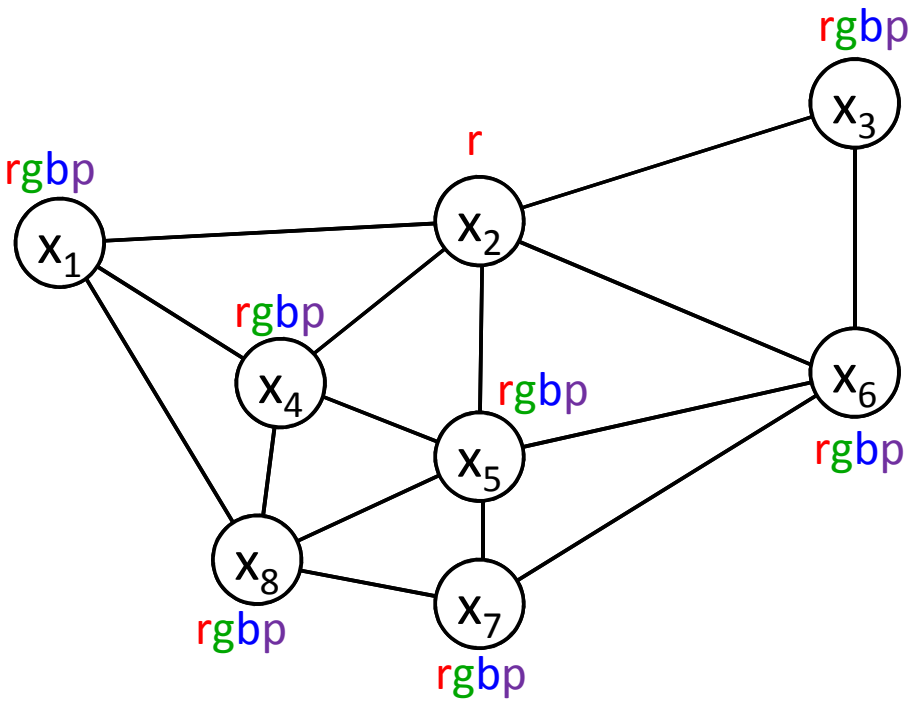
Constraint propagation: can we remove inconsistent domain values?

29



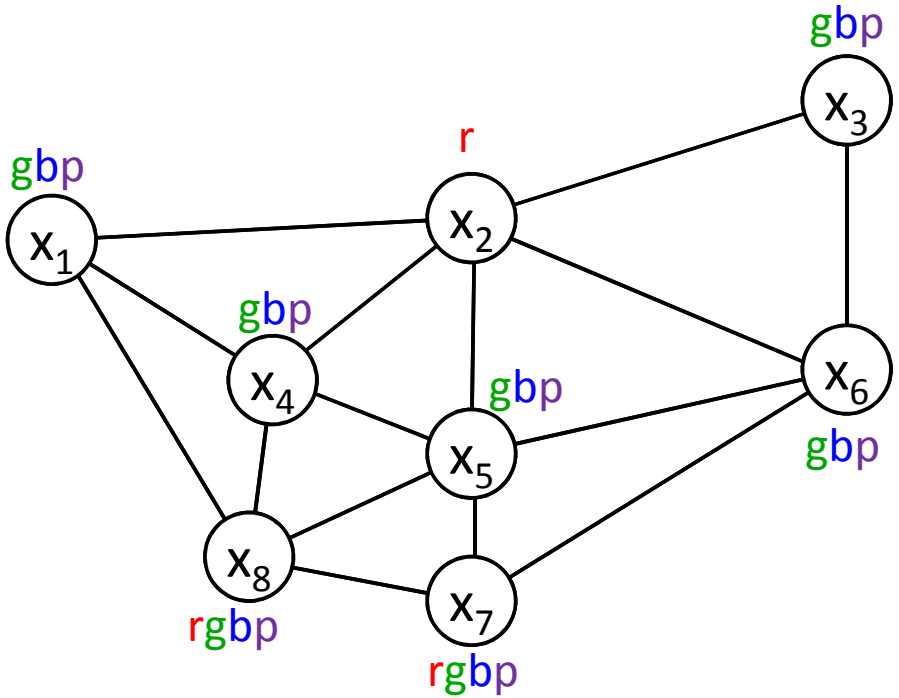
Search: guess a value for a variable
(but be prepared to backtrack): $x_2 = r$

30



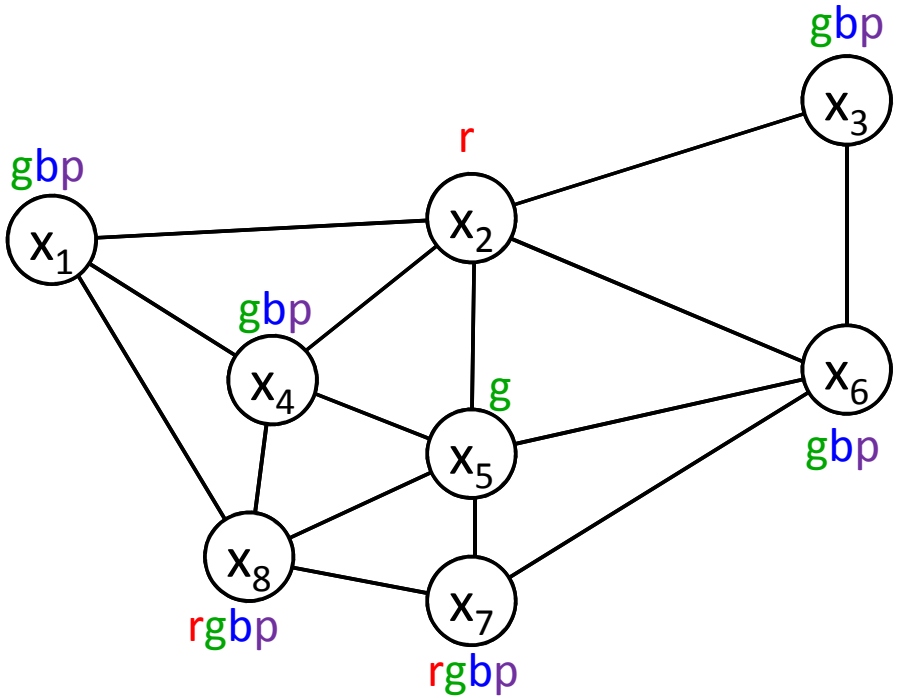
Constraint propagation

31



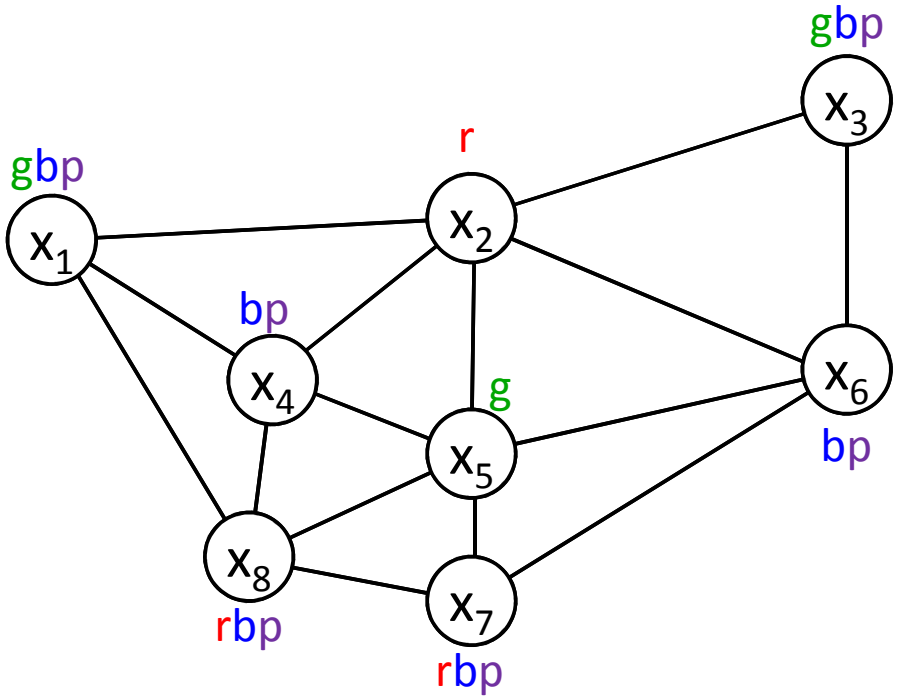
Search: guess a value for a variable
(but be prepared to backtrack): $x_5 = g$

32



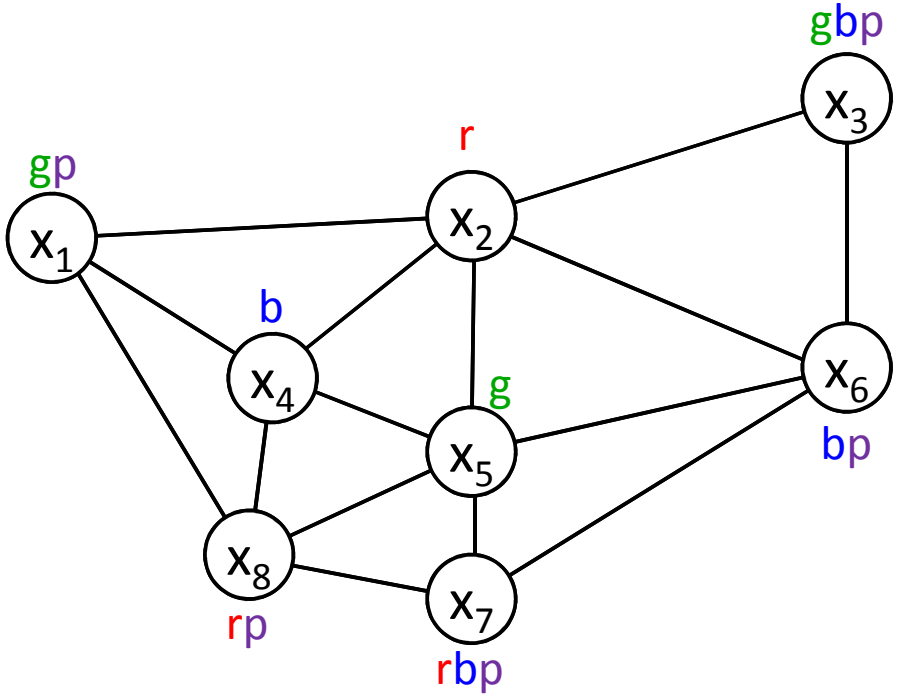
Constraint propagation

33



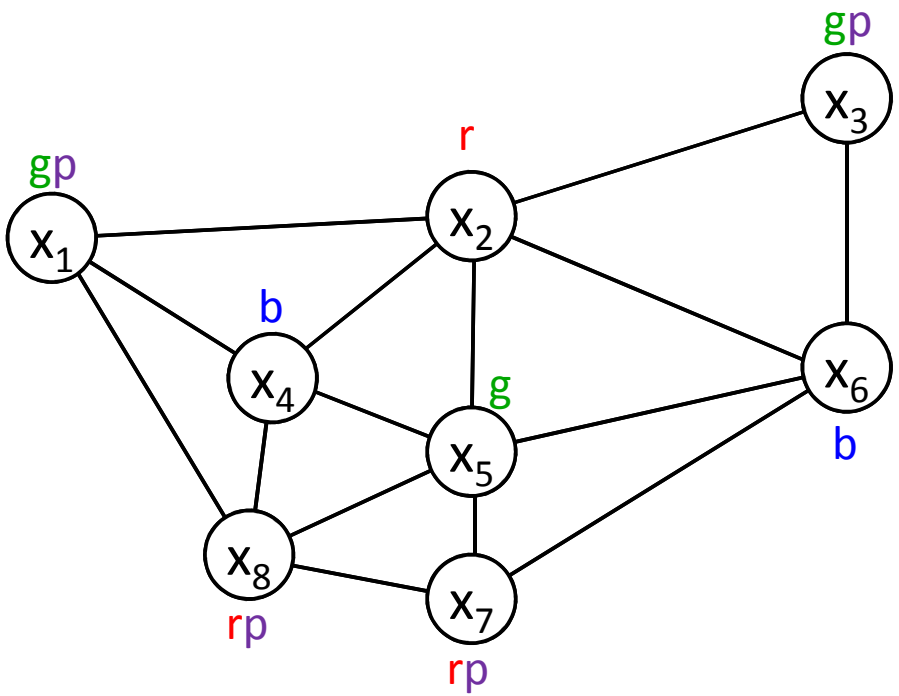
Search: guess a value for a variable
(but be prepared to backtrack): $x_4 = b$

34

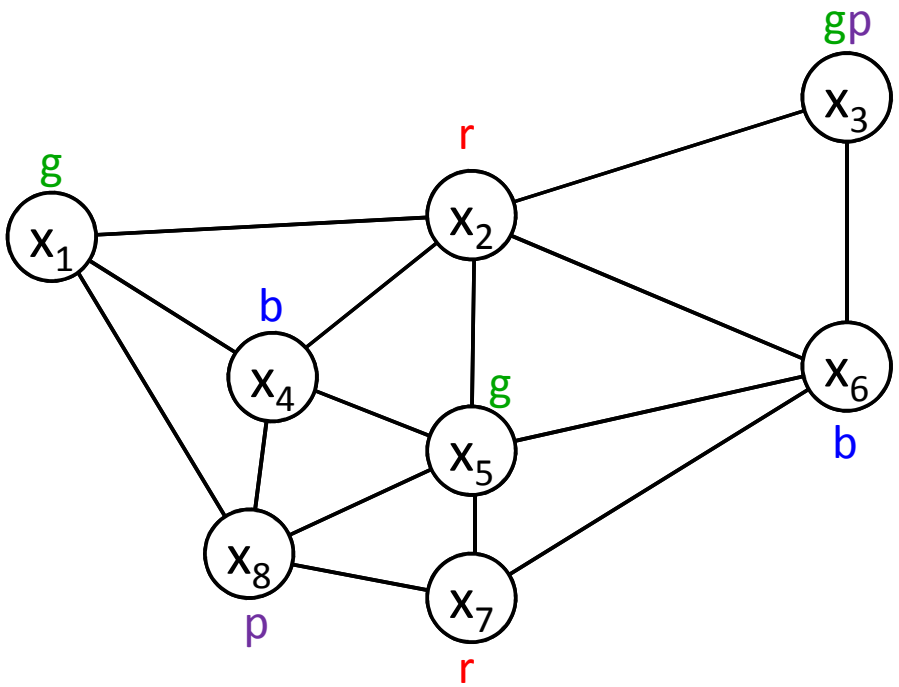


Next search choice: $x_6 = b$

35

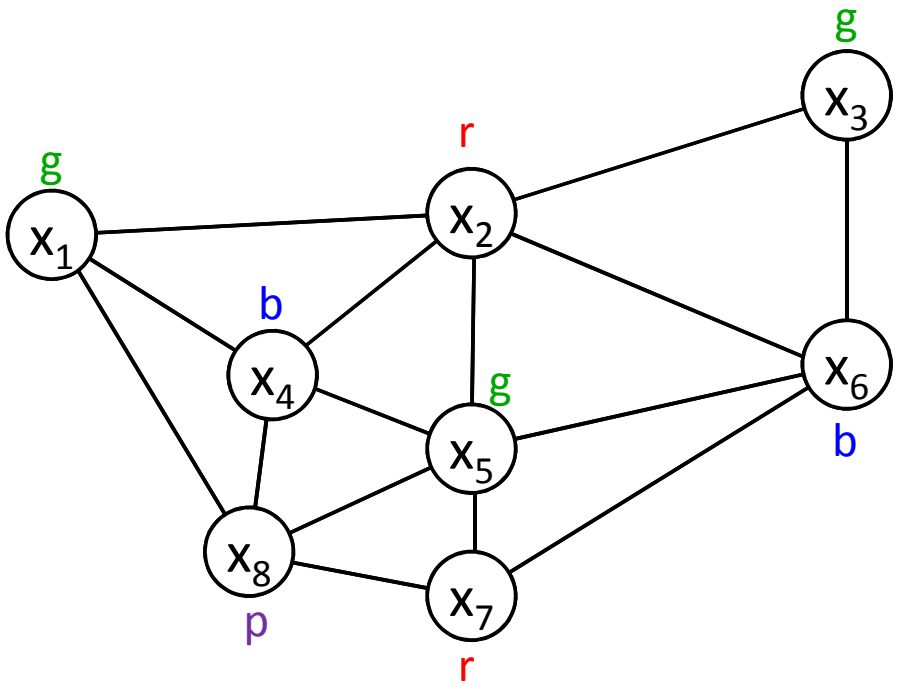


Next search choice: x_8 = p



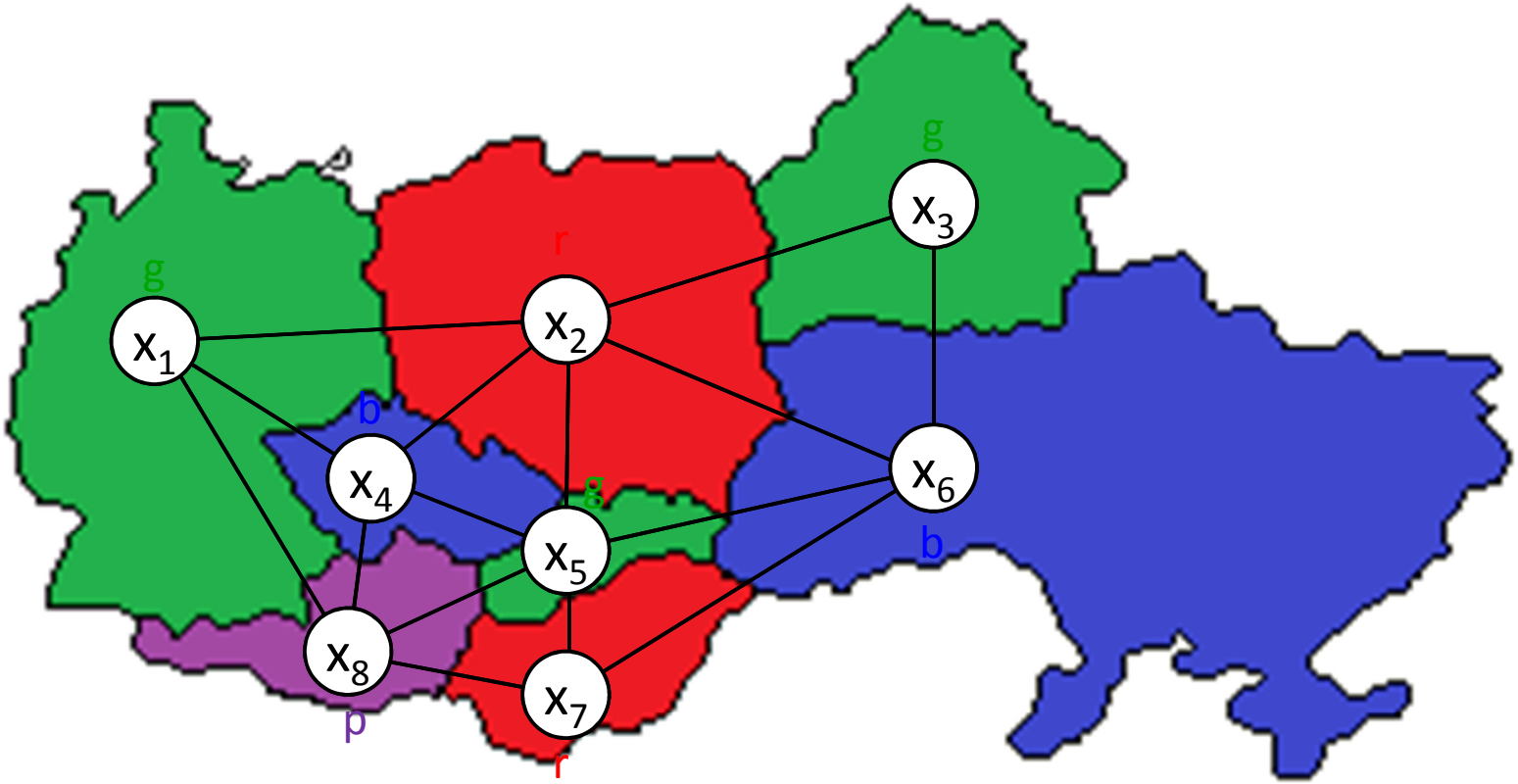
Next search choice: $x_3 = g$

37



We found a solution!

38



39

Given a set of variables, each taking a value from a domain of possible values, find an assignment to all variables that satisfy the constraints.

- Variables:



- Domain:



- Constraints: Adjacent states must be colored differently

Find a solution!



_ _ _ _ _ _ _



R _ _ _ _ _ _



R G _ _ _ _ _



R G R _ _ _ _

⋮ ⋮

R G R R R R R

How many leaf nodes are expanded in the worst case?



_ _ _ _ _ _ _



R _ _ _ _ _ _



R G _ _ _ _ _



R G R _ _ _ _

⋮ ⋮

R G R R R R R

How many leaf nodes are expanded in the worst case? $3^7 = 2187$



_ _ _ _ _ _ _



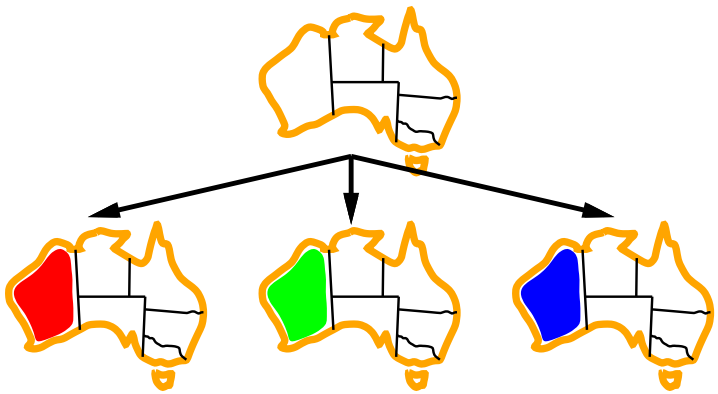
R _ _ _ _ _ _

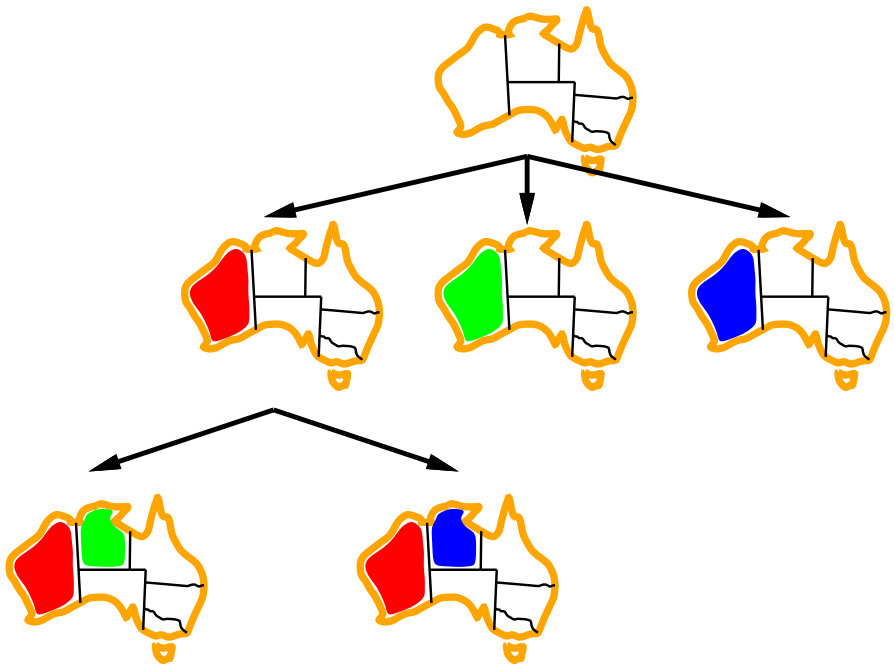
This is bad.
How can we improve it?

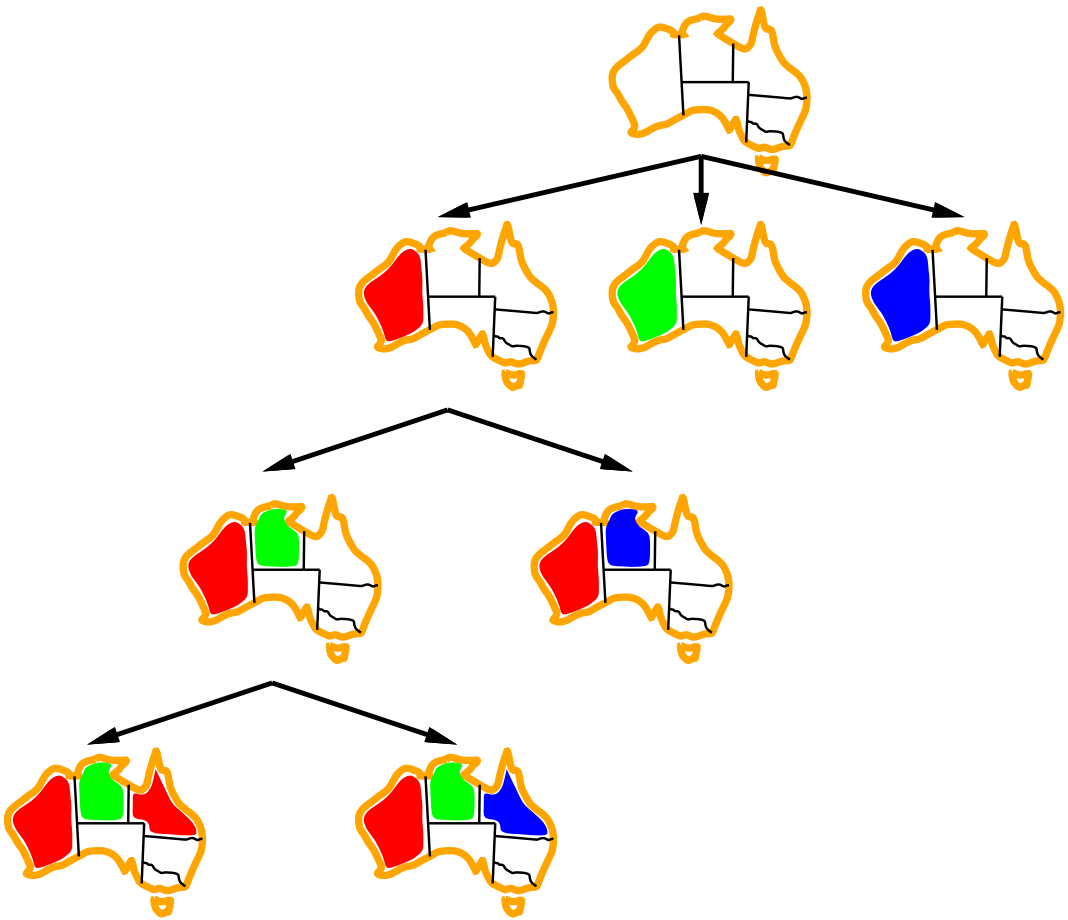
R G R R R R R

How many leaf nodes are expanded in the worst case? $3^7 = 2187$









Constraint Programming

Given:

- a set X of variables, $X = x_1, \ldots x_n$;
- a set D of domains $D_1, \ldots, D_n$, associated with $x_1, \ldots, x_n$, respectively;
- a set C of constraints, $c_j \subseteq D(x_{i_1}) \times \cdots \times D(x_{i_k})$, i.e., relations specifying the acceptable combination of values among the variables.

Definition (CSP)

A Constraint Satisfaction Problem (CSP) is the problem to find an assignment $x_i := d_i \in D_i$, such that all the constraints are satisfied.

In the same settings as CSPs, given in addition:

- an objective function $f : D_1 \times \ldots \times D_n \rightarrow \mathbb{R}$

Definition (COP)

A *Constrained Optimization Problem* (COP) is the problem to find the assignment $x_i := d_i \in D_i$, such that all the constraints are satisfied and the cost function f is (w.l.o.g.) minimized.

- It is both **procedural** and **declarative**.
  - procedural = write a computer program
  - declarative = state constraints on the solution

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  - declarative = state constraints on the solution
- It uses **global constraints** to exploit problem structure:
  - global constraint = constraint that contains many simpler constraints

- It is both **procedural** and **declarative**.
  - procedural = write a computer program
  - declarative = state constraints on the solution
- It uses **global constraints** to exploit problem structure:
  - global constraint = constraint that contains many simpler constraints
- It uses **filtering** and **constraint propagation** to reduce the search space.
  - filtering = reduce variable domains
  - propagation = pass domains to next constraint

- Example: solve this:

$$3x_1 + x_2 + x_3 = 10$$
$$x_1, x_2, x_3 \text{ pairwise distinct}$$
$$x_1, x_2 \in \{1,2\}, x_3 \in \{1,2,3\}$$

Note that $x_1 = x_2 = x_3 = 2$ is not allowed.

- Example: solve this:

$$3x_1 + x_2 + x_3 = 10$$
$$x_1, x_2, x_3 \text{ pairwise distinct}$$
$$x_1, x_2 \in \{1,2\}, x_3 \in \{1,2,3\}$$

- Purely procedural model:

For x_1 = 1,2:
  For x_2 = 1,2:
    If $x_1 \neq x_2$ then
      For x_3 = 1,2,3:
        If $x_1 \neq x_3$ and $x_2 \neq x_3$ then
          If $3x_1 + x_2 + x_3 = 10$ then print x_1, x_2, x_3

- Example: solve this:

$$3x_1 + x_2 + x_3 = 10$$
$$x_1, x_2, x_3 \text{ pairwise distinct}$$
$$x_1, x_2 \in \{1,2\}, x_3 \in \{1,2,3\}$$

- Purely declarative model:

$$3x_1 + x_2 + x_3 = 10$$
$$x_1 \neq x_2$$
$$x_1 \neq x_3$$
$$x_2 \neq x_3$$
$$x_1, x_2 \in \{1,2\}, x_3 \in \{1,2,3\}$$

- Example: solve this:

$$3x_1 + x_2 + x_3 = 10$$
$$x_1, x_2, x_3 \text{ pairwise distinct}$$
$$x_1, x_2 \in \{1,2\}, x_3 \in \{1,2,3\}$$

- CP model:

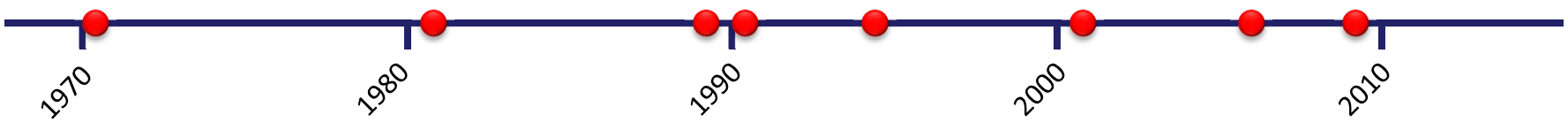
$$3x_1 + x_2 + x_3 = 10$$
$$\text{alldiff}(x_1, x_2, x_3)$$
$$x_1, x_2 \in \{1,2\}, x_3 \in \{1,2,3\}$$

This **global constraint** (all-different) enforces $x_1 \neq x_2$, $x_1 \neq x_3$, $x_2 \neq x_3$.

Constraint Programming = model (representation) +
propagation (reasoning, inference) +
search (reasoning, inference)


Modeling and Solving COPs
basic solution methods
propagation
consistency
(tree) search
advanced solution methods
hybrid
heuristic
basics
search
constraints
(types of) domains
sets
intervals
finite domains
CP concepts
modeling
global constraints
redundant constraints
symmetry elimination
special-purpose constraints
modeling of optimization problems
problem reduction
software
GECODE
COMET
...
examples
timetabling
scheduling
sudoku
crypto-arithmetic
n-queens

- 1970s: Image processing applications in AI; Search+qualitative inference
- 1980s: Logic Programming (Prolog); Search + logical inference
- 1989: CHIP System; Constraint Logic Programming
- 1990s: Constraint Programming; Industrial Solvers (ILOG, Eclipse,…)
- 1994: Advanced inference for *alldifferent* and *resource scheduling*
- 2000s: Global constraints; integrated methods; modeling languages
- 2006: CISCO Systems acquires Eclipse CLP solver
- 2009: IBM acquires ILOG CP Solver & Cplex



"CP represents one of the closest approaches computer science has yet made to the Holy Grail of programming: the user states the problem, the computer solves it."

.

Eugene C. Freuder, CONSTRAINTS, April 1997

Commercial

• IBM ILOG CP Optimizer	C++ library; free for academic users
• Kalis (Artelys / Fico)	C++ library
• SICSTUS-Prolog	Prolog-based library
• CHIP V5 (Cosytec)	C++/Prolog-based library
• Comet (Dynadec)	C++ library; free for academic users
• SCIP [1]	C library; free for academic users

[1] SCIP actually solves 'constraint integer programs' and combines MIP/LP solving with CP and SAT techniques

Non-commercial (in terms of licensing)

- Gecode — C++ library
- Eclipse — Prolog-based library
- Google OR-tools — C++ library
- Choco — Java library
- JaCoP — Java library
- Minion — black-box solver; specific problem input
- Mistral — C++ library

Examples

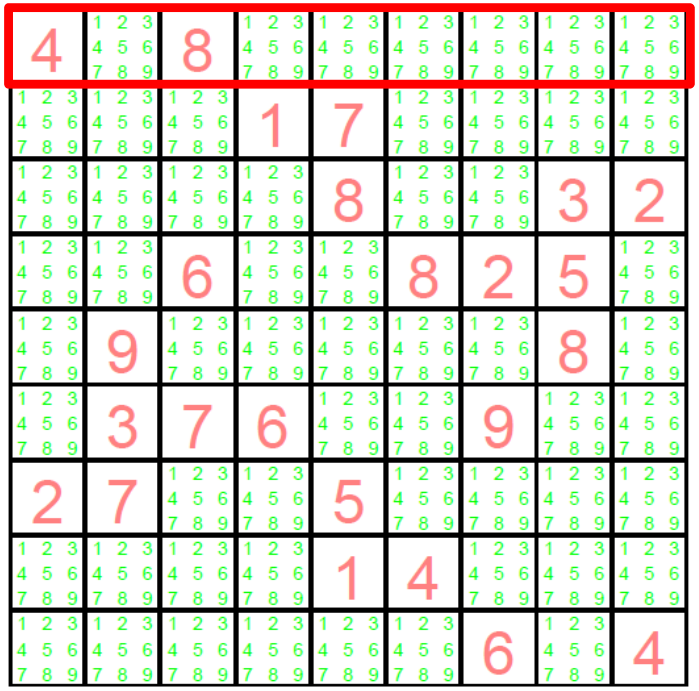
Sudoku

4		8						
			1	7				
				8			3	2
		6			8	2	5	
	9						8	
	3	7	6			9		
2	7			5				
					1	4		
						6		4

Fill blanks with numbers1-9.

Thanks to Helmut Simonis for this example.

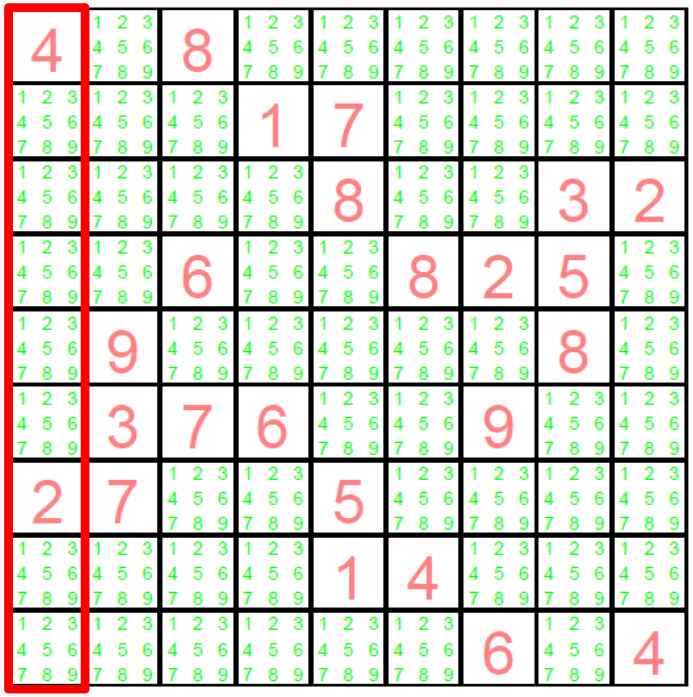
Sudoku



Fill blanks with numbers1-9.

Numbers all different in each row,

Sudoku

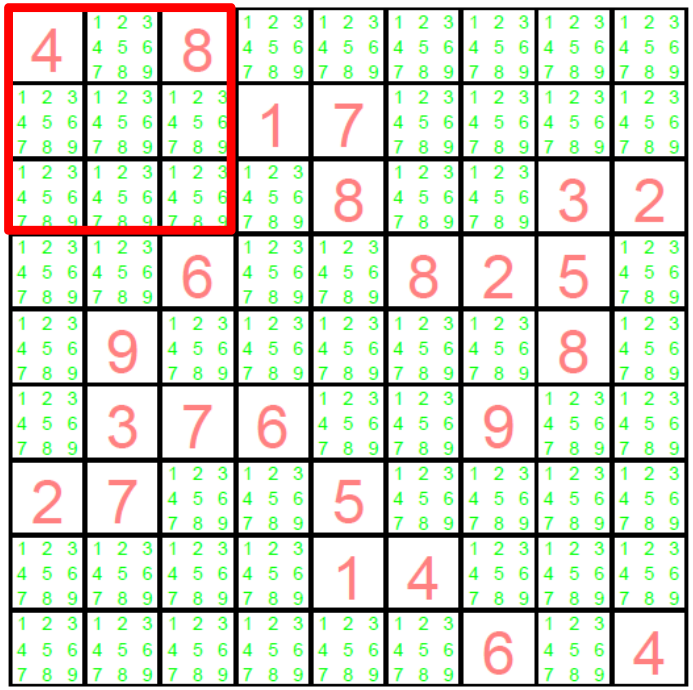


Fill blanks with numbers1-9.

Numbers all different in each row,

In each column,

Sudoku



Fill blanks with numbers1-9.

Numbers all different in each row,

In each column,

And in each 3x3 square.

Sudoku

4		8						
			1	7				
				8			3	2
		6			8	2	5	
	9						8	
	3	7	6			9		
2	7			5				
					1	4		
						6		4

Let x_{ij} = number in cell i,j

Sudoku

4		8						
			1	7				
				8			3	2
		6			8	2	5	
	9						8	
	3	7	6			9		
2	7			5				
					1	4		
						6		4

Let x_{ij} = number in cell i,j

alldiff(x_{11}, …, x_{19})

Sudoku

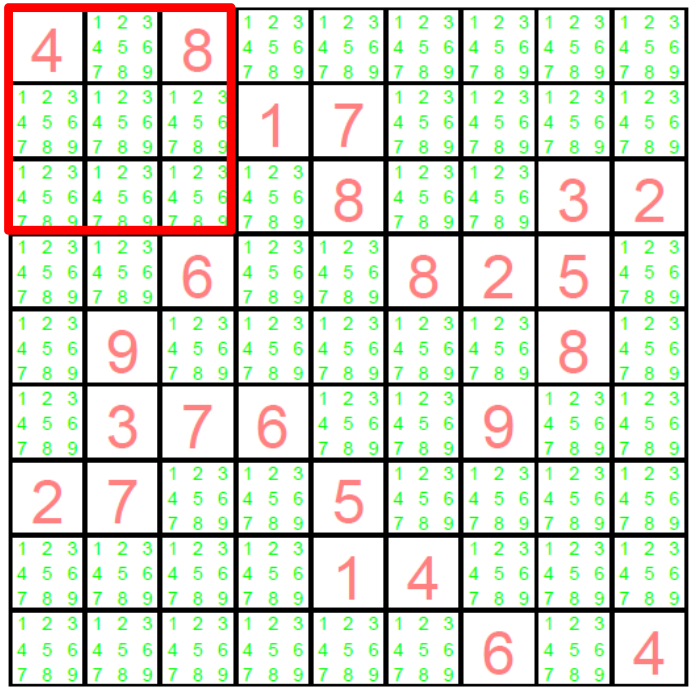
4		8						
			1	7				
				8			3	2
		6			8	2	5	
	9						8	
	3	7	6			9		
2	7			5				
					1	4		
						6		4

Let x_{ij} = number in cell i,j

alldiff(x_{11}, …, x_{19})

alldiff(x_{11}, …, x_{91})

Sudoku



Let x_{ij} = number in cell i,j

alldiff(x_{11}, …, x_{19})

alldiff(x_{11}, …, x_{91})

alldiff(x_{11},x_{12},x_{13},x_{21},x_{22},x_{23},x_{31},x_{32},x_{33})

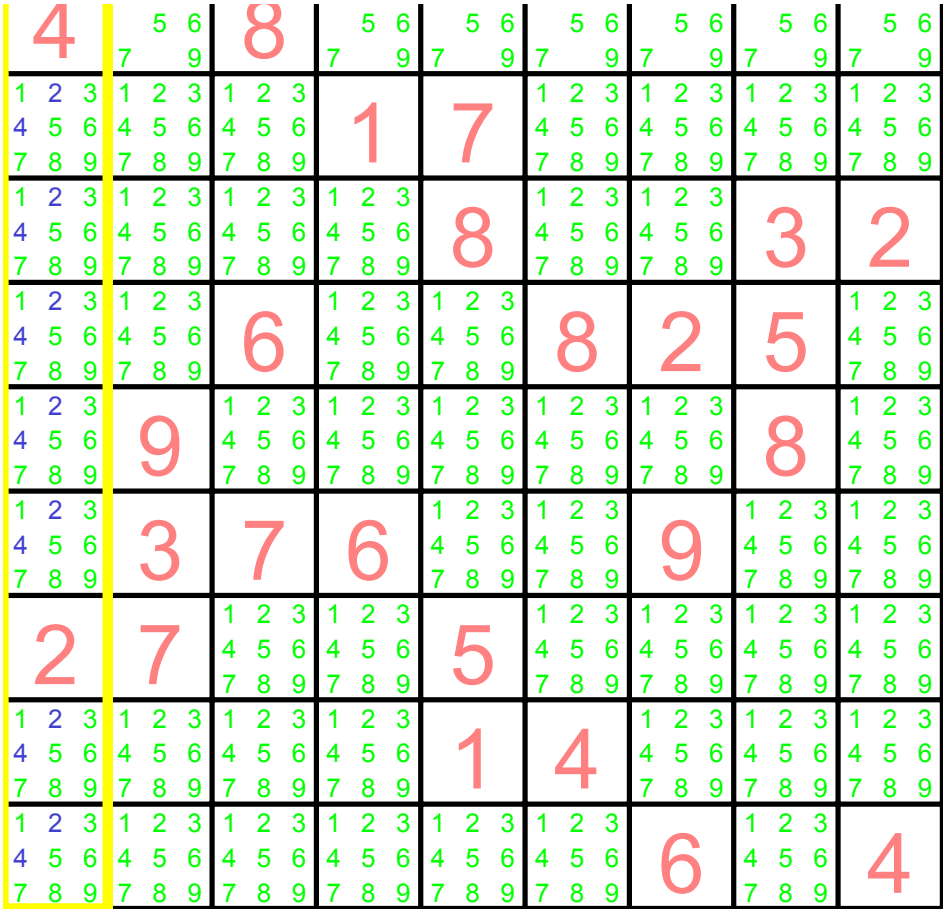
etc.

4	4 5 6 7 8 9	8	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1	7	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	8	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	3	2
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	6	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	8	2	5	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	8	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	3	7	6	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
2	7	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	5	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1	4	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	6	1 2 3 4 5 6 7 8 9	4

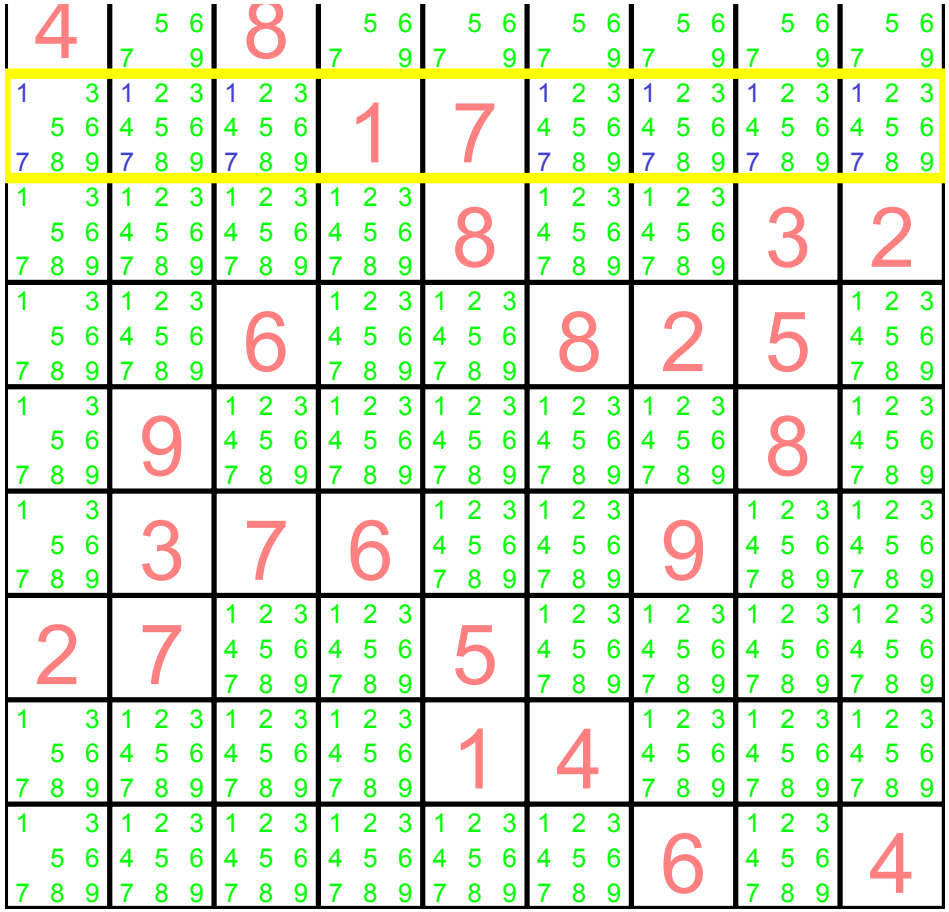
Skip Animation

4	4 5 6 7 8 9	8	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9	4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1	7	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	8	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	3	2
1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	6	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	8	2	5	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	8	1 2 3 4 5 6 7 8 9
1 2 3 4 5 6 7 8 9	3	7	6	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
2	7	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	5	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9
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1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	1 2 3 4 5 6 7 8 9	6	1 2 3 4 5 6 7 8 9	4

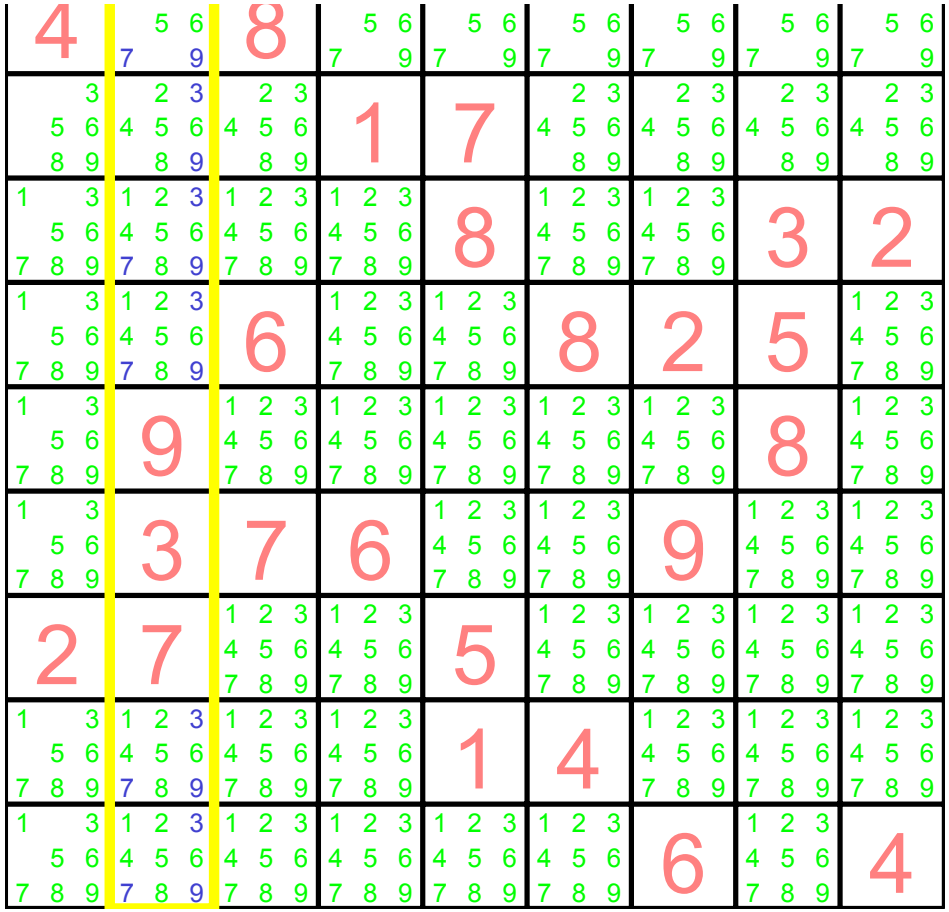
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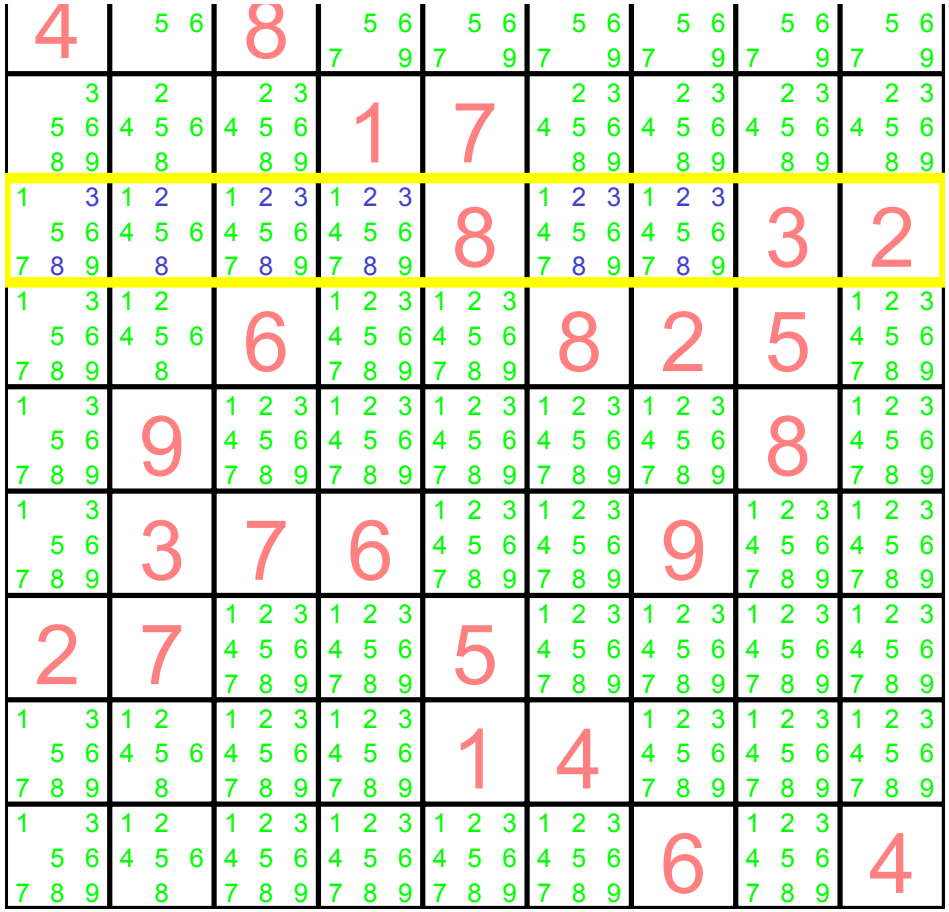


◂ Back to Start ▸▸ Skip Animation

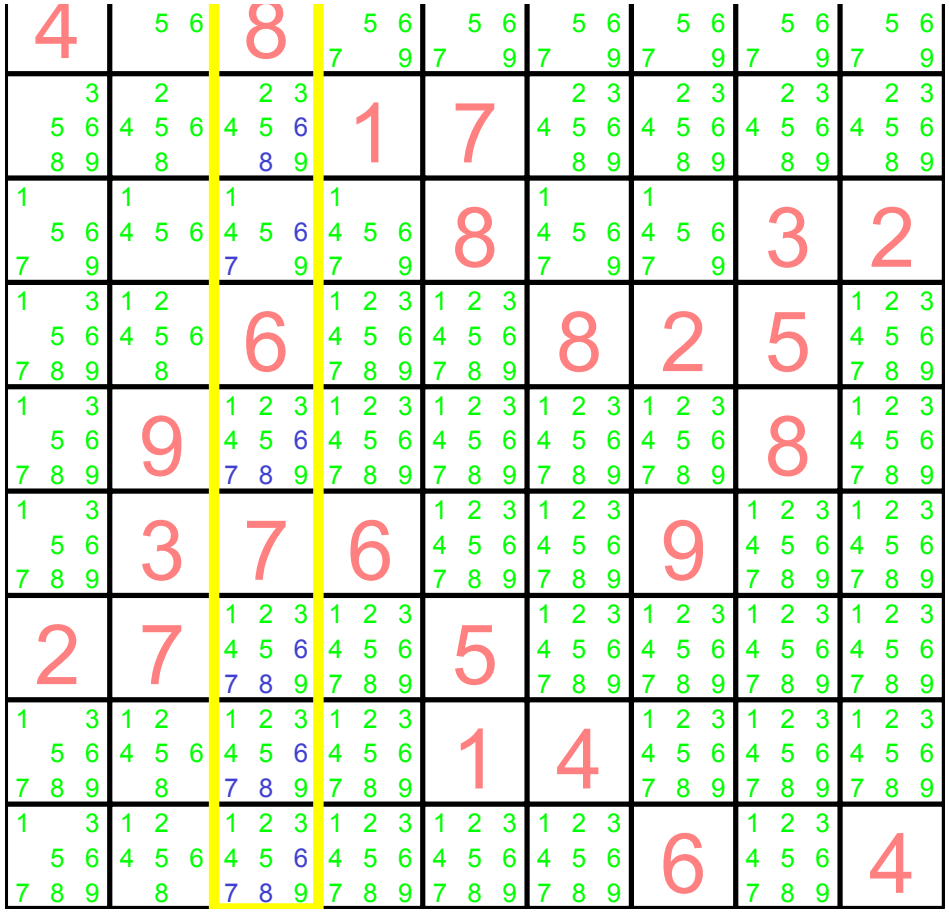


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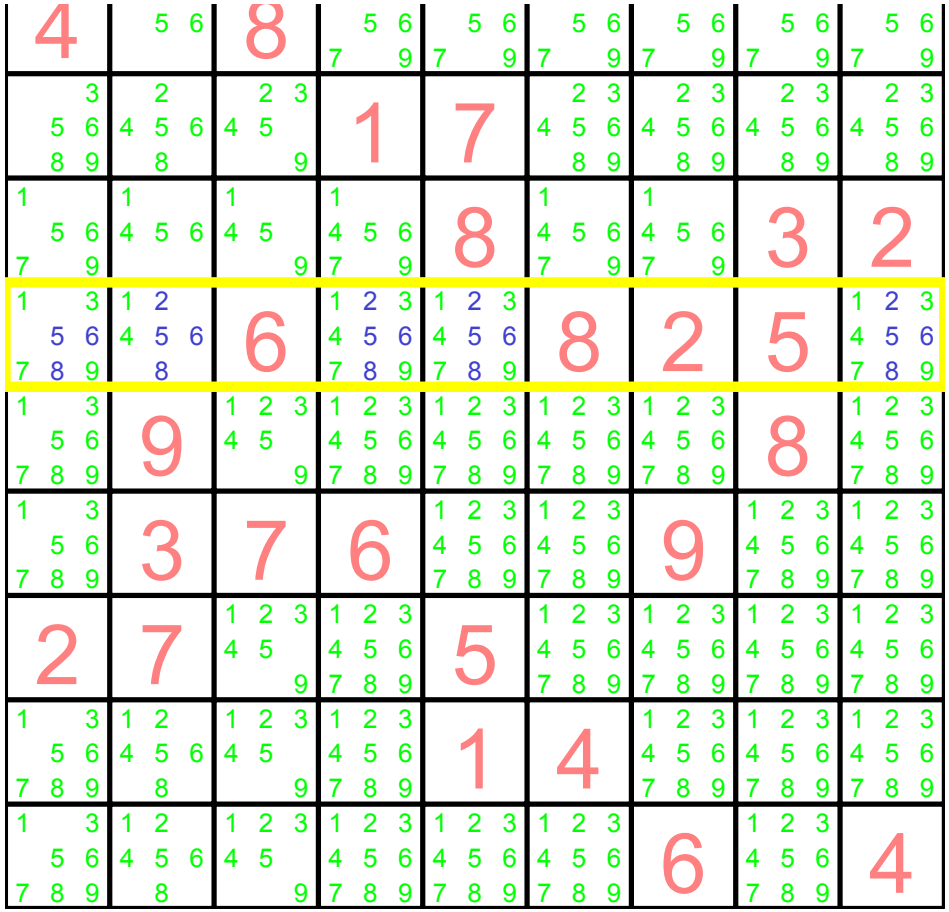

4 | 8 | 1 | 7 | 8 | 3 | 2 | 6 | 9 | 2 | 7 | 5 | 1 | 4 | 6 | 9 | 3 | 4



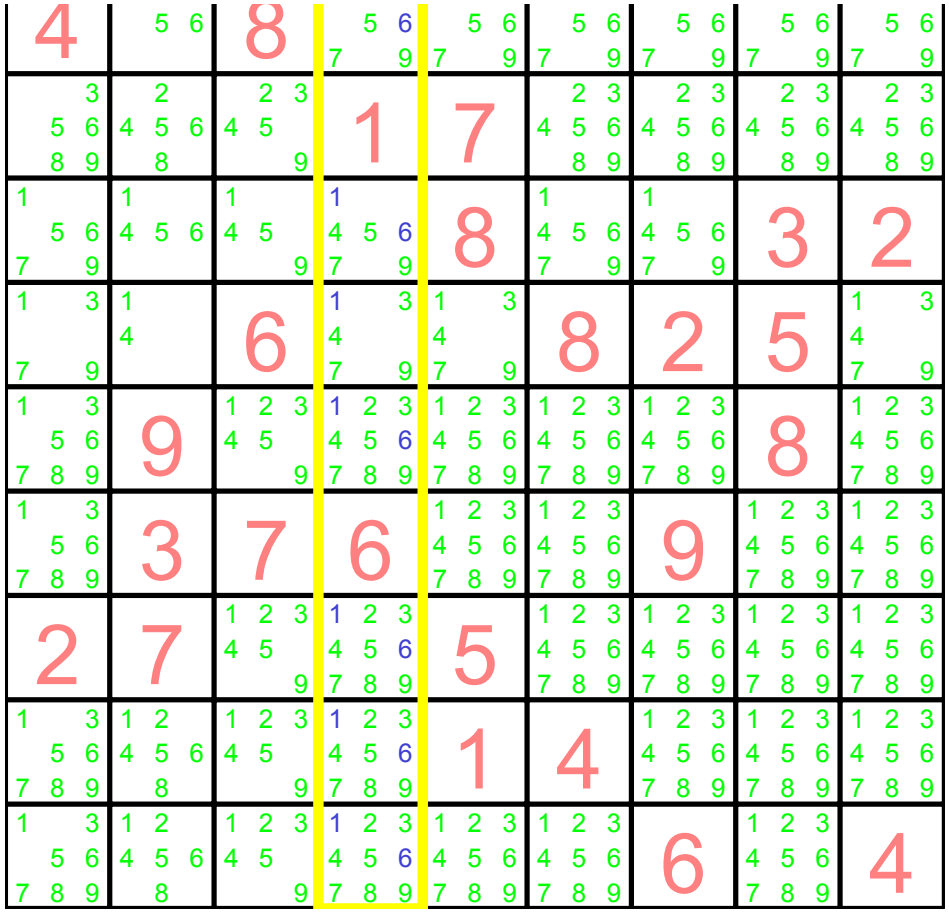
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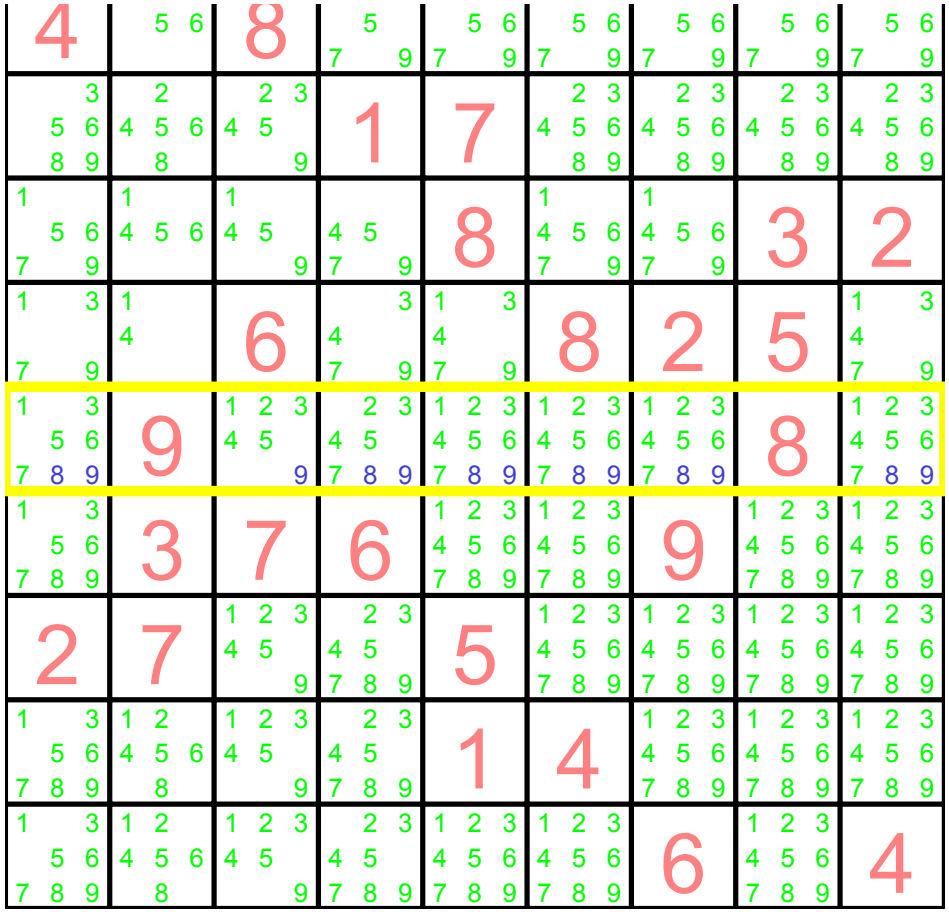
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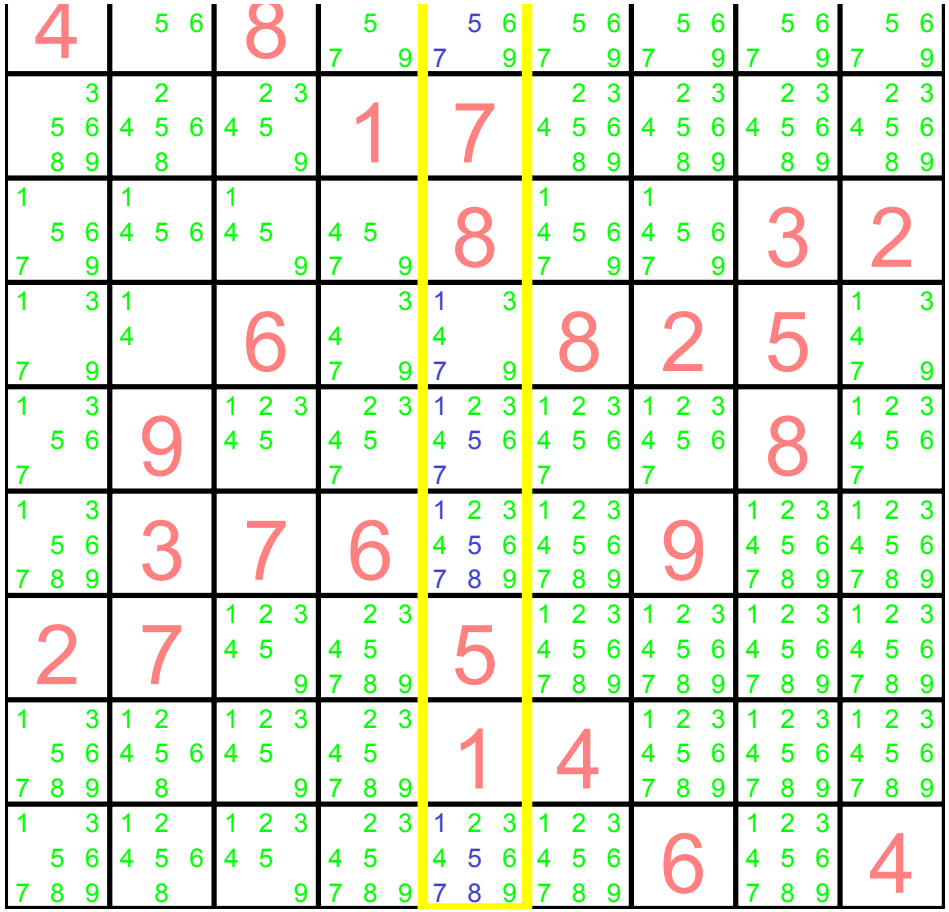
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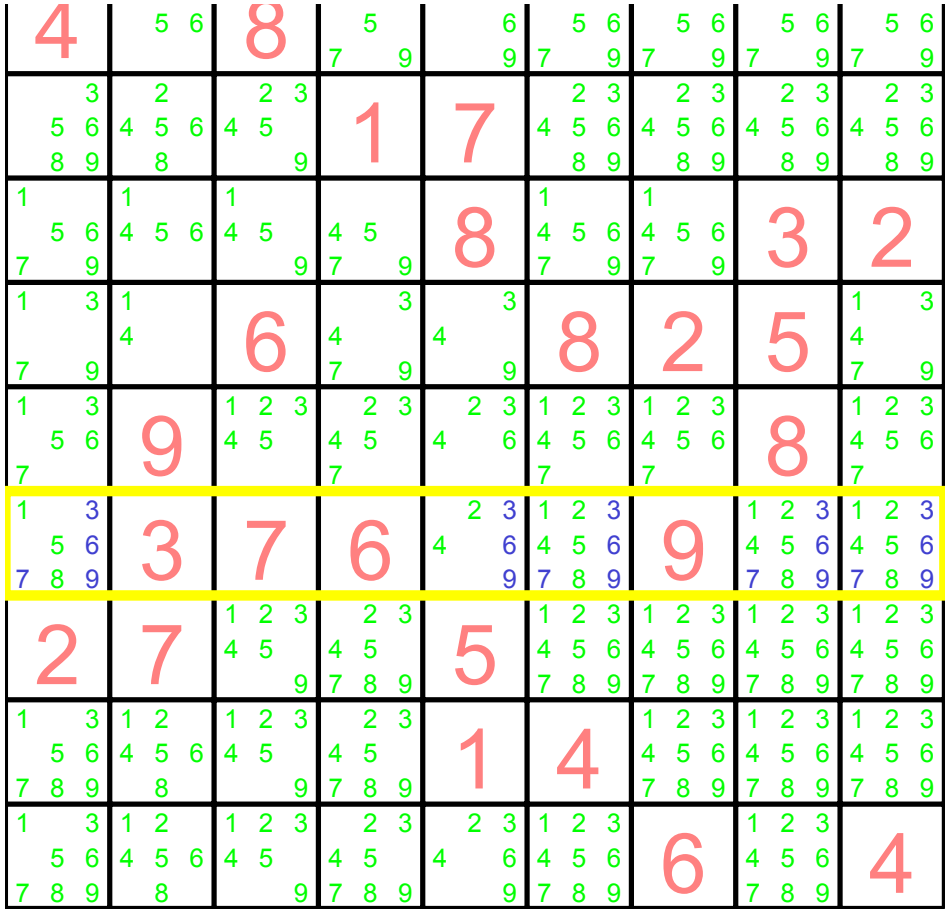
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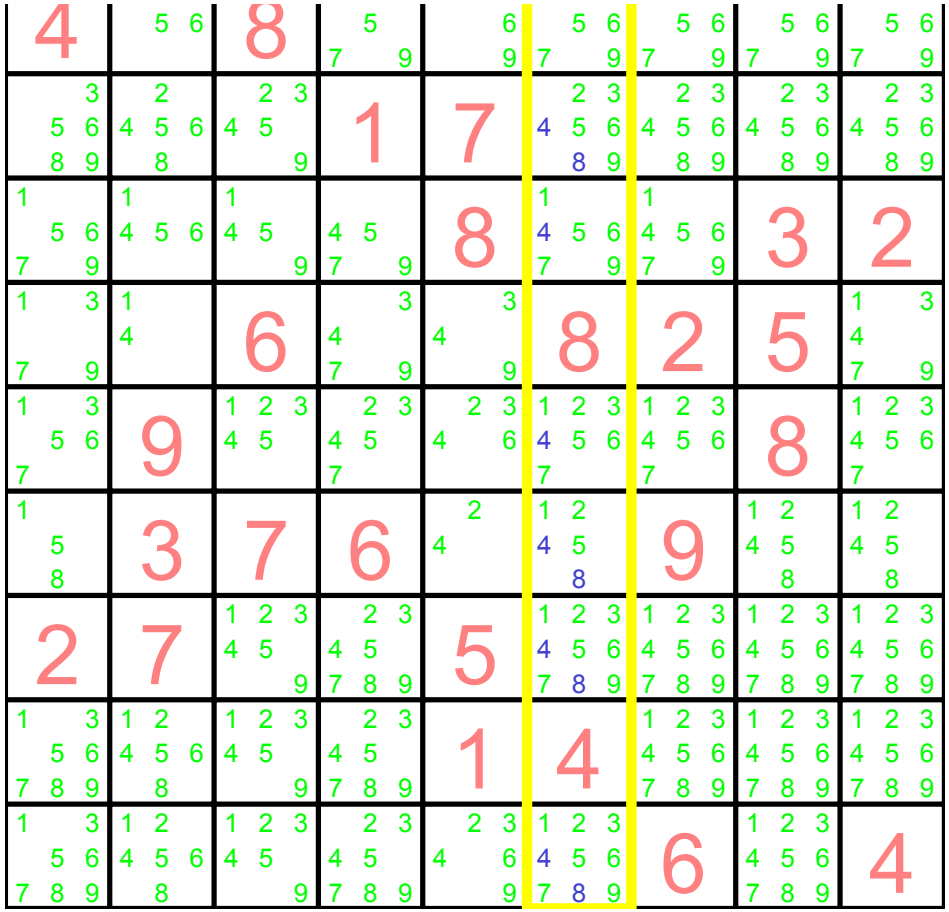
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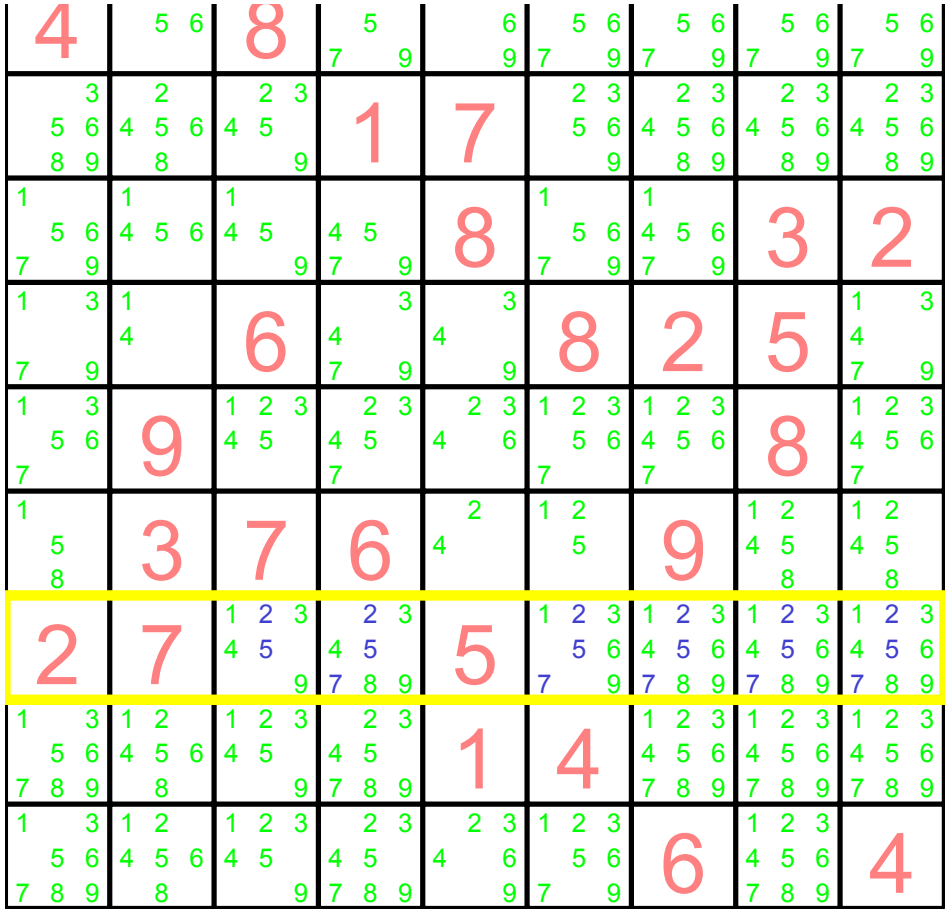
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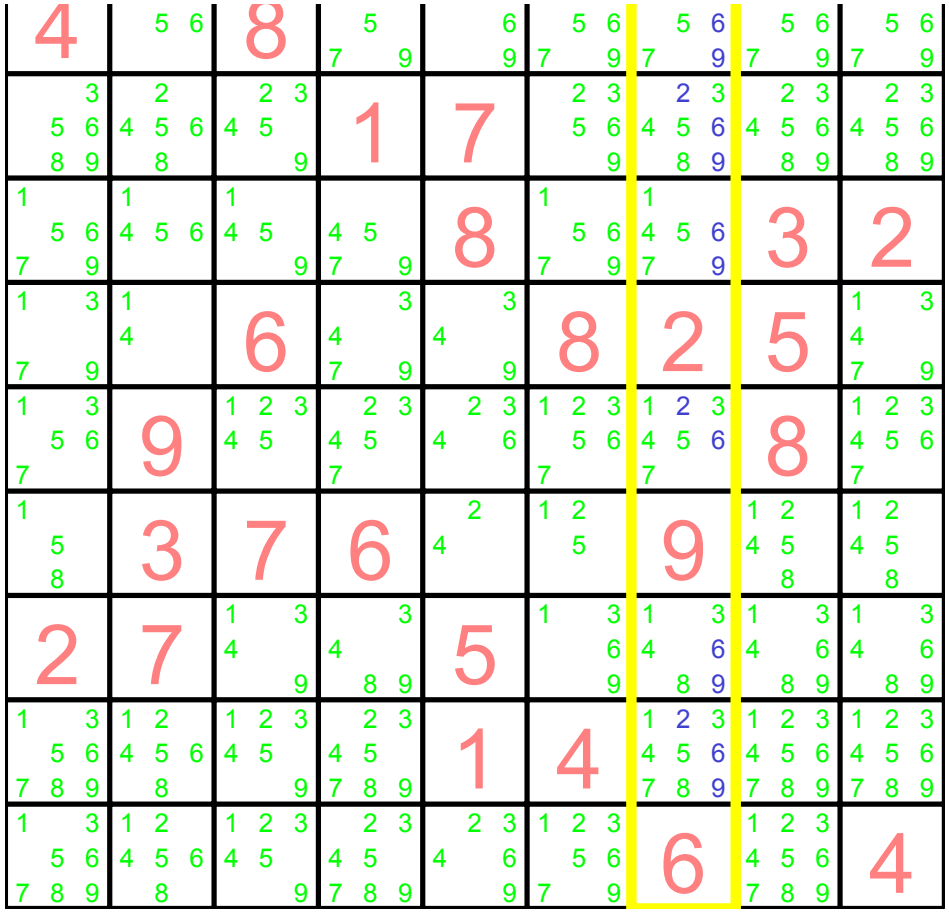
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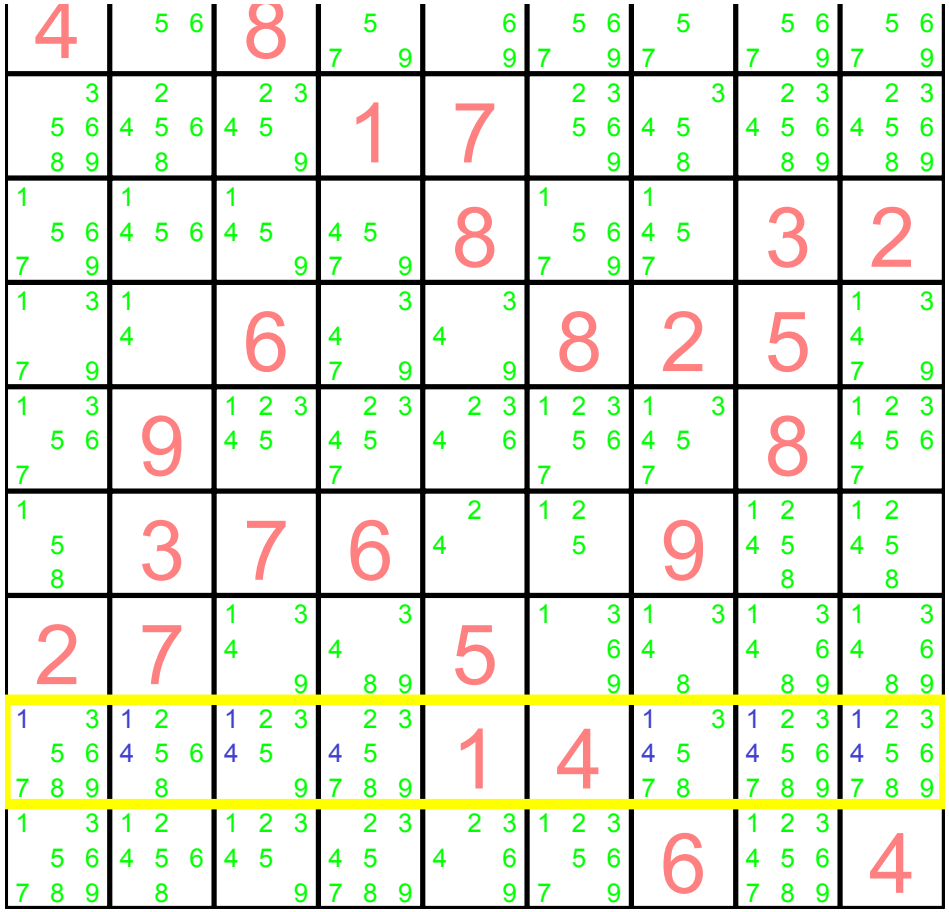
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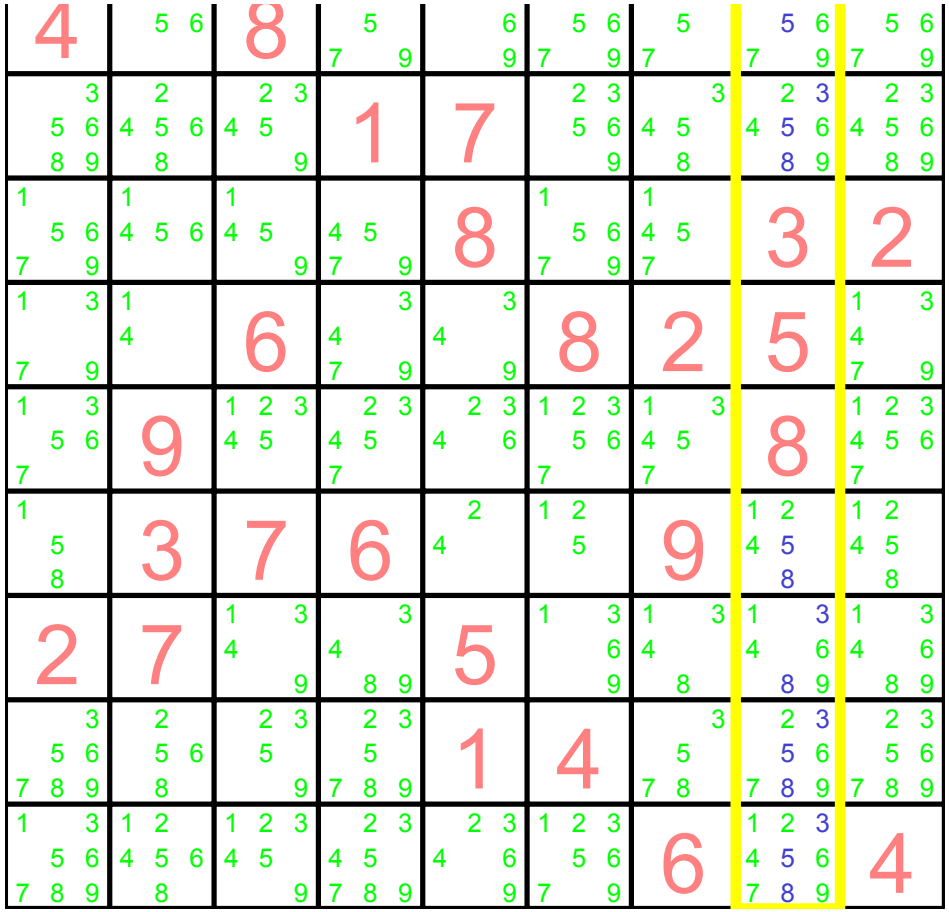
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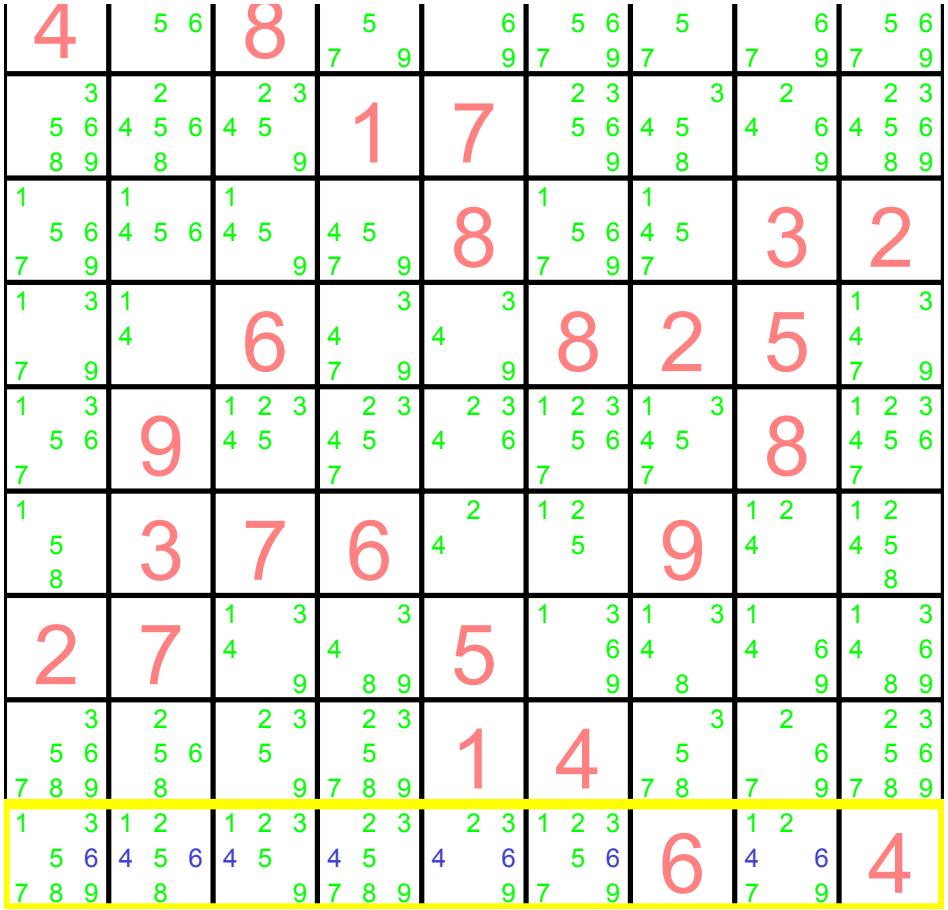
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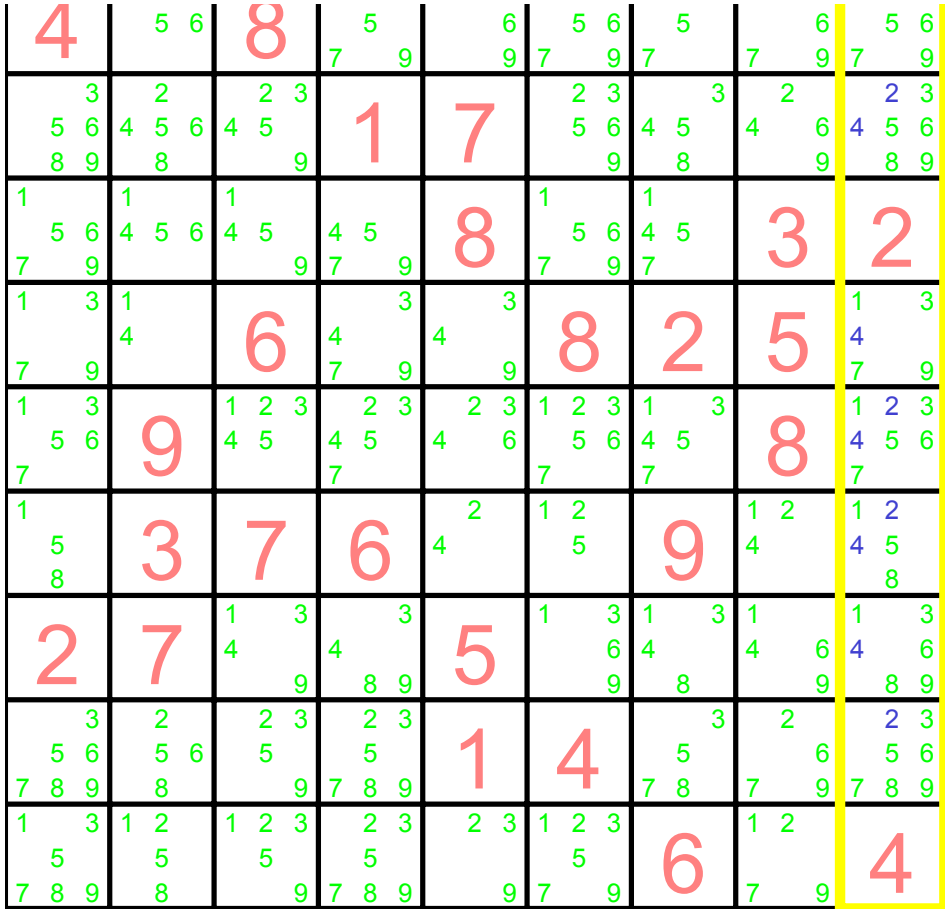
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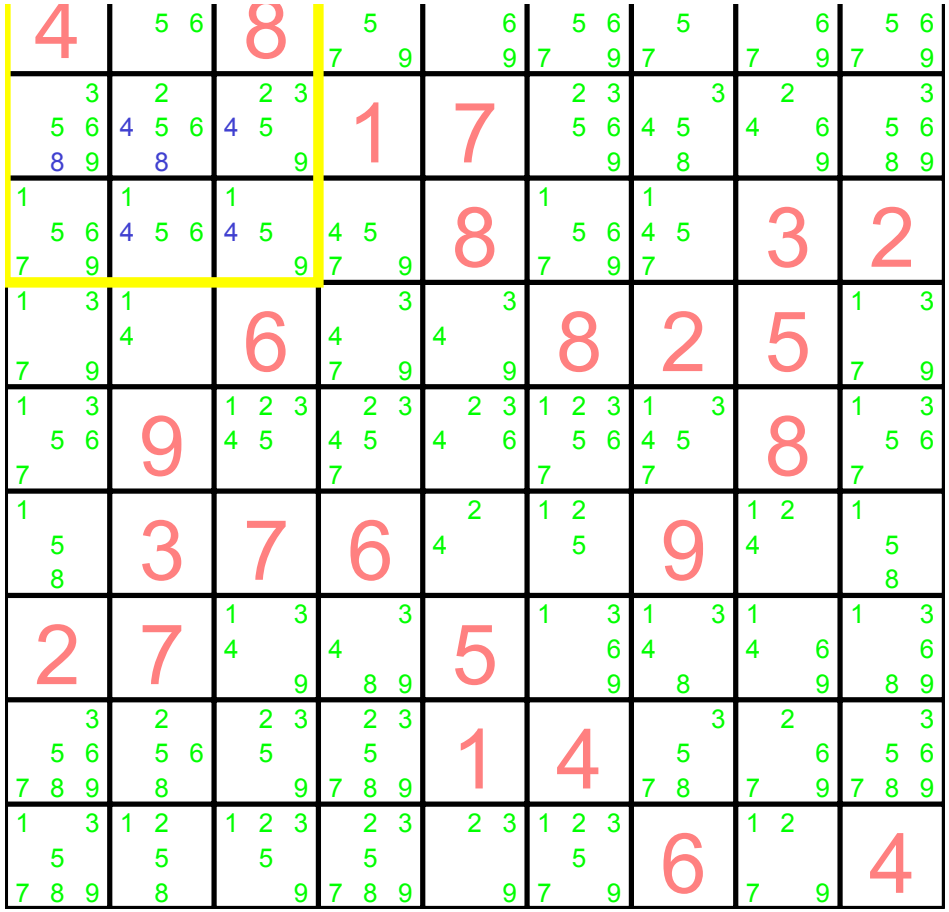
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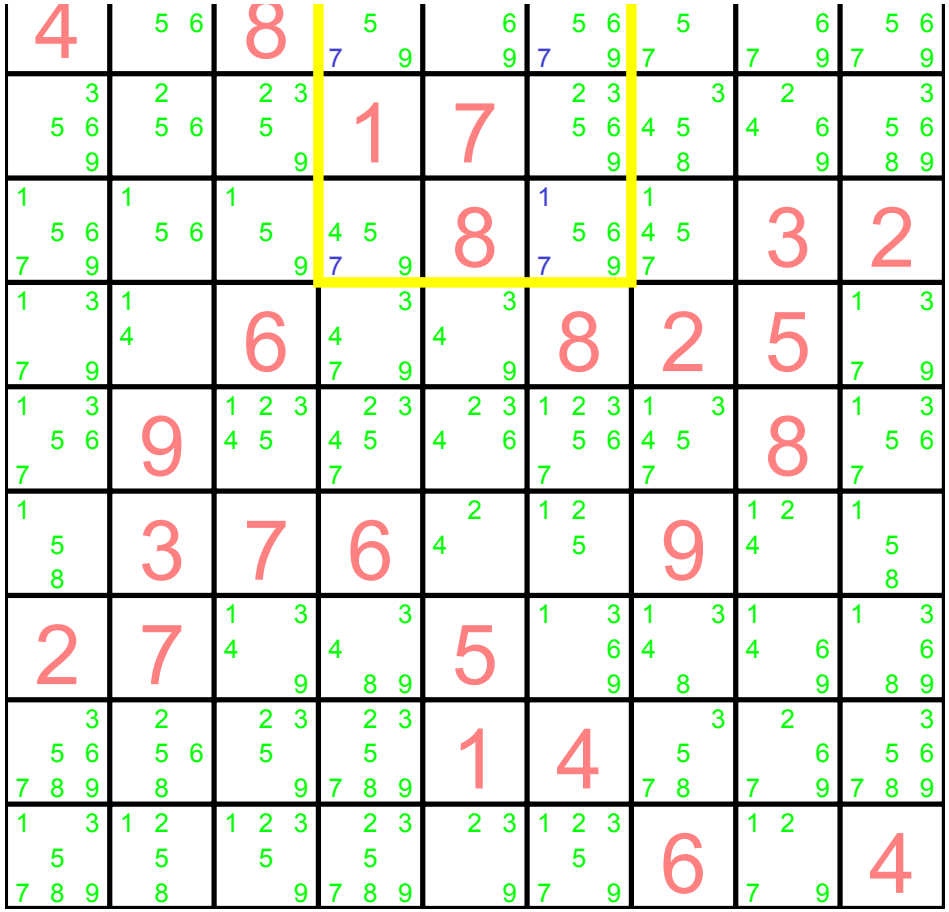
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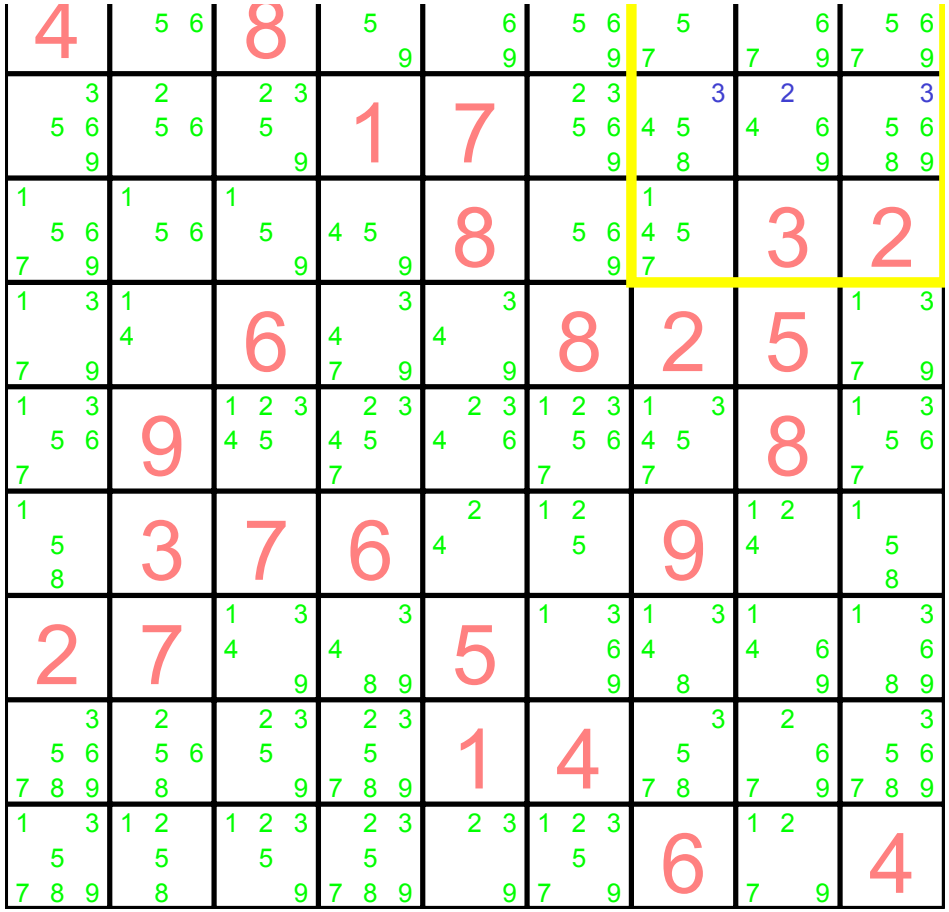
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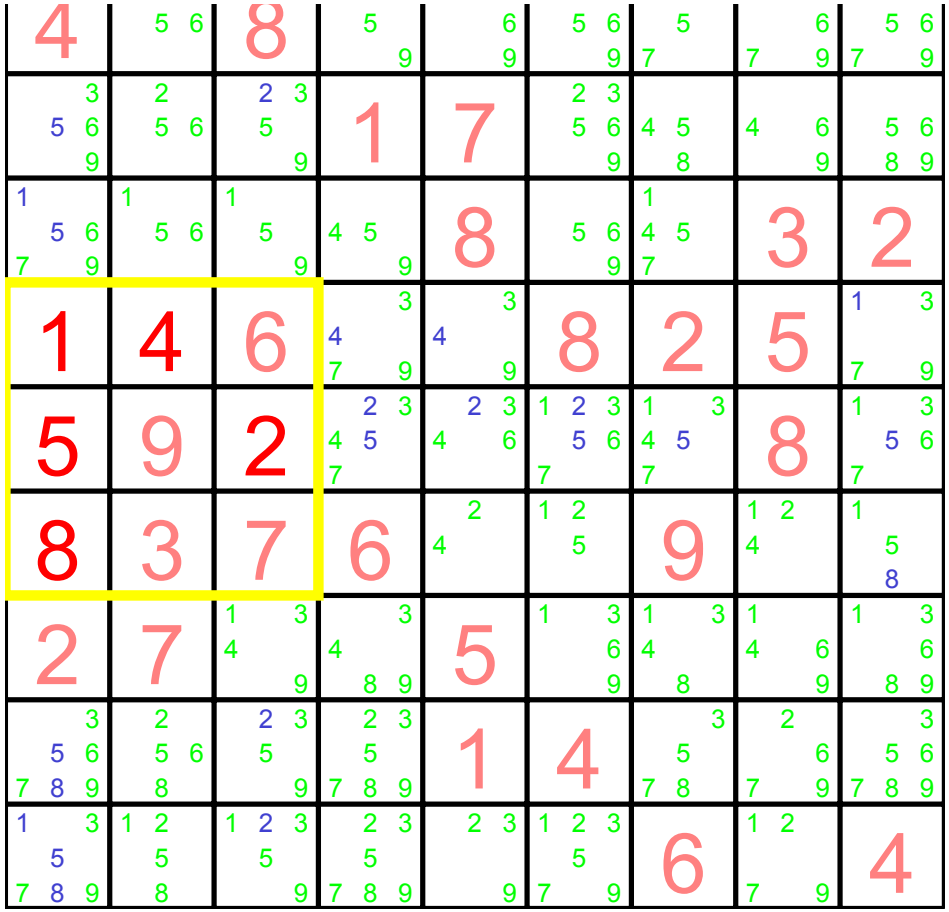
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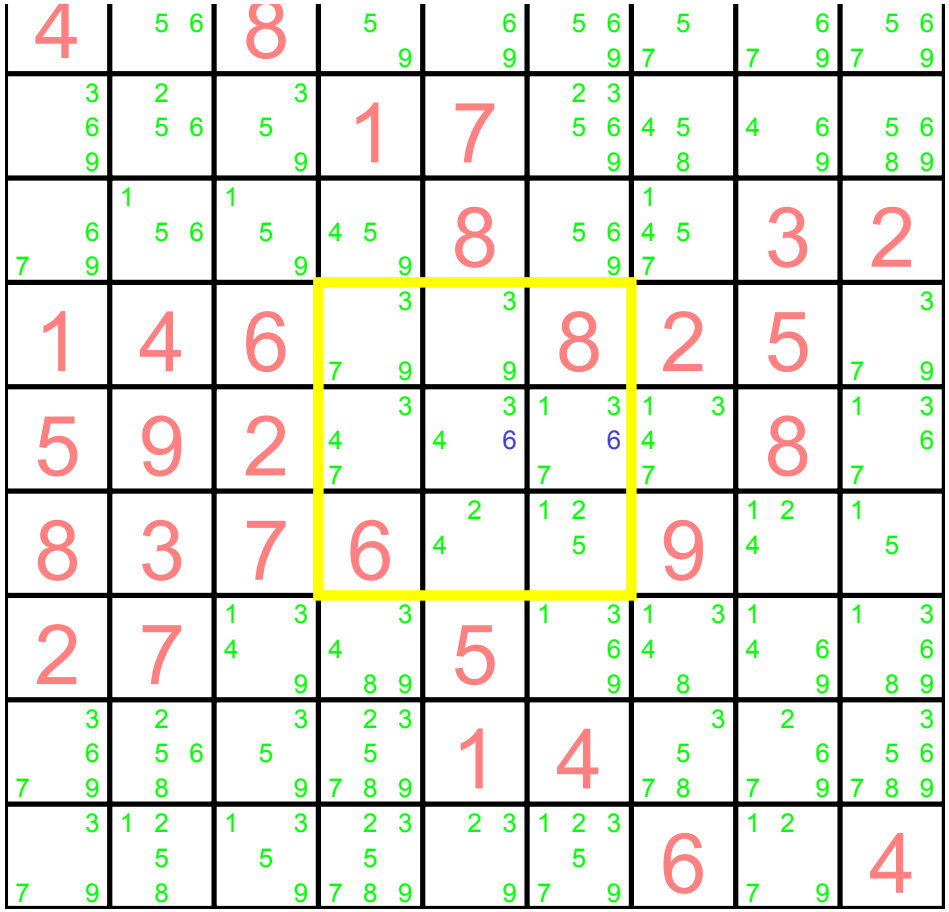
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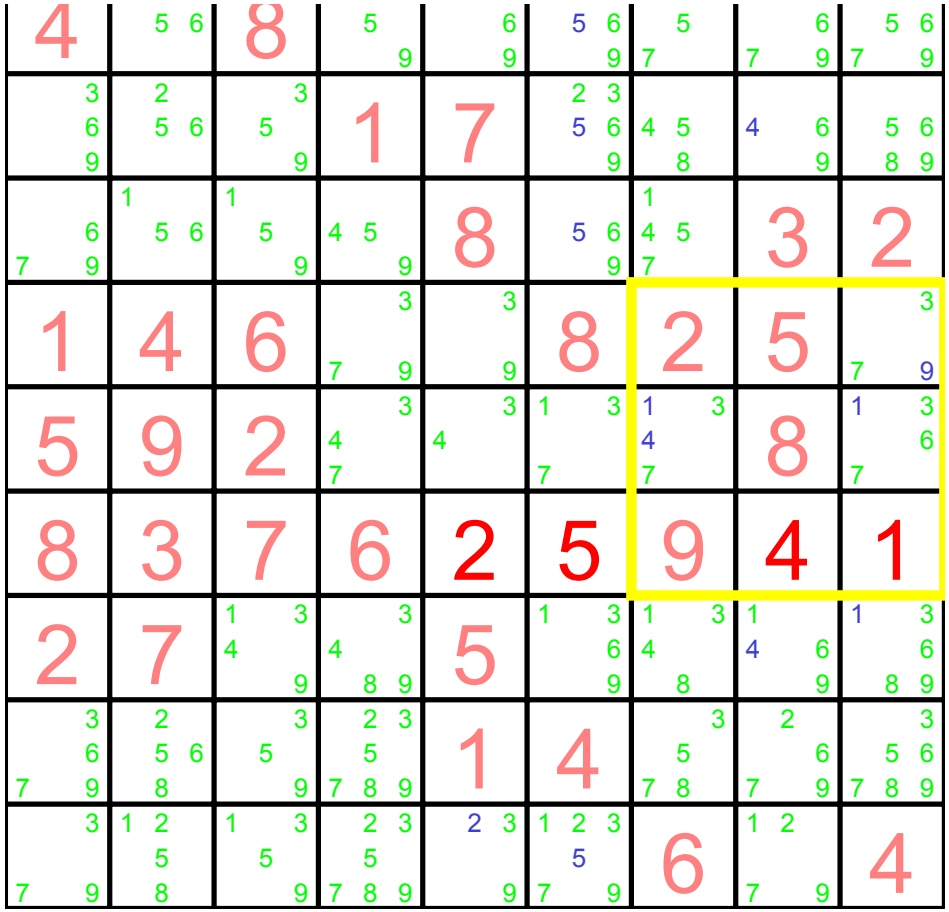
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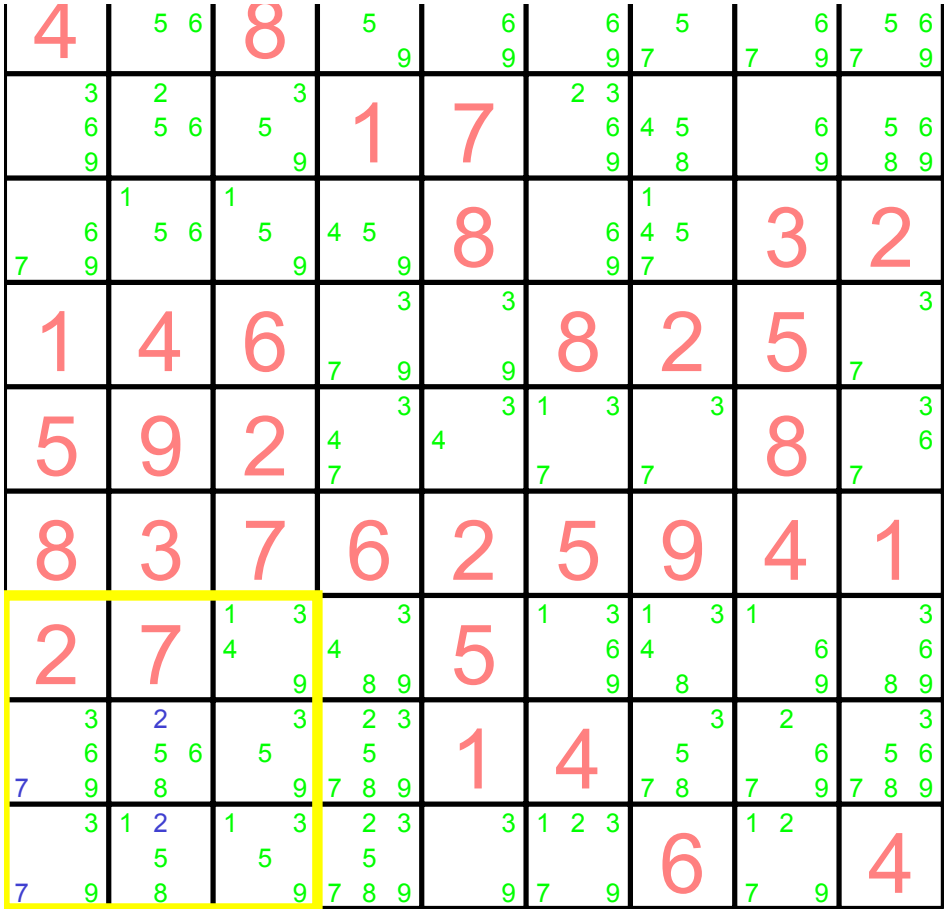
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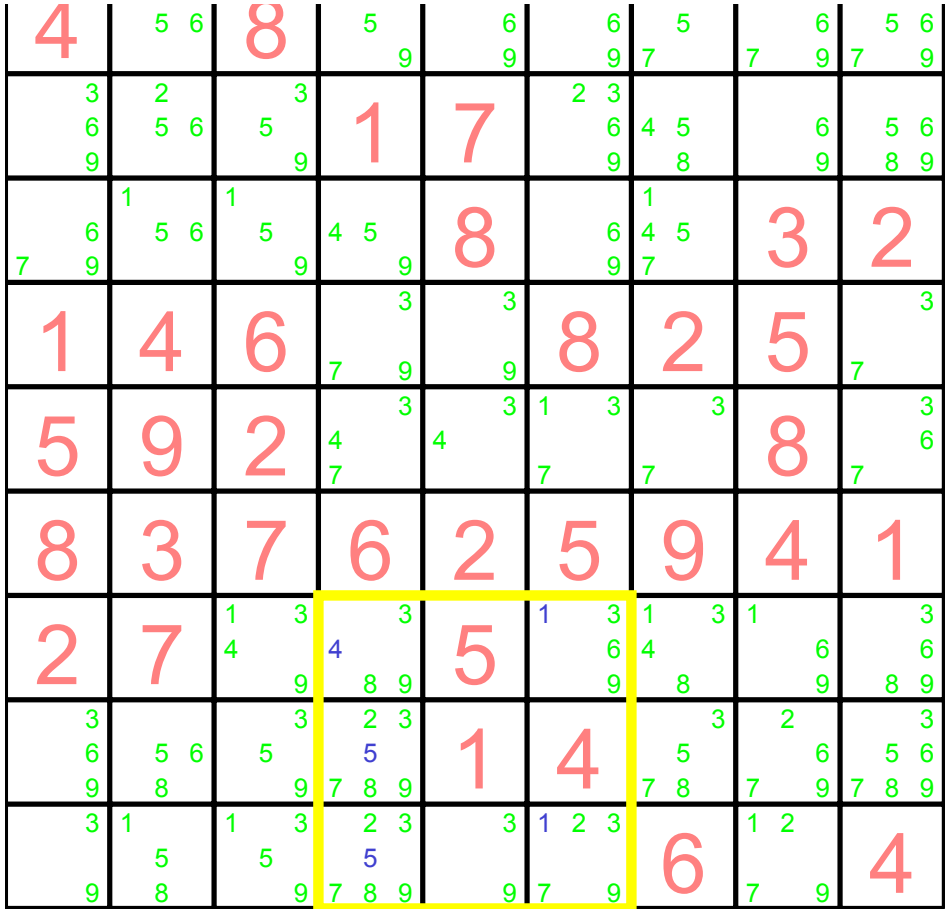
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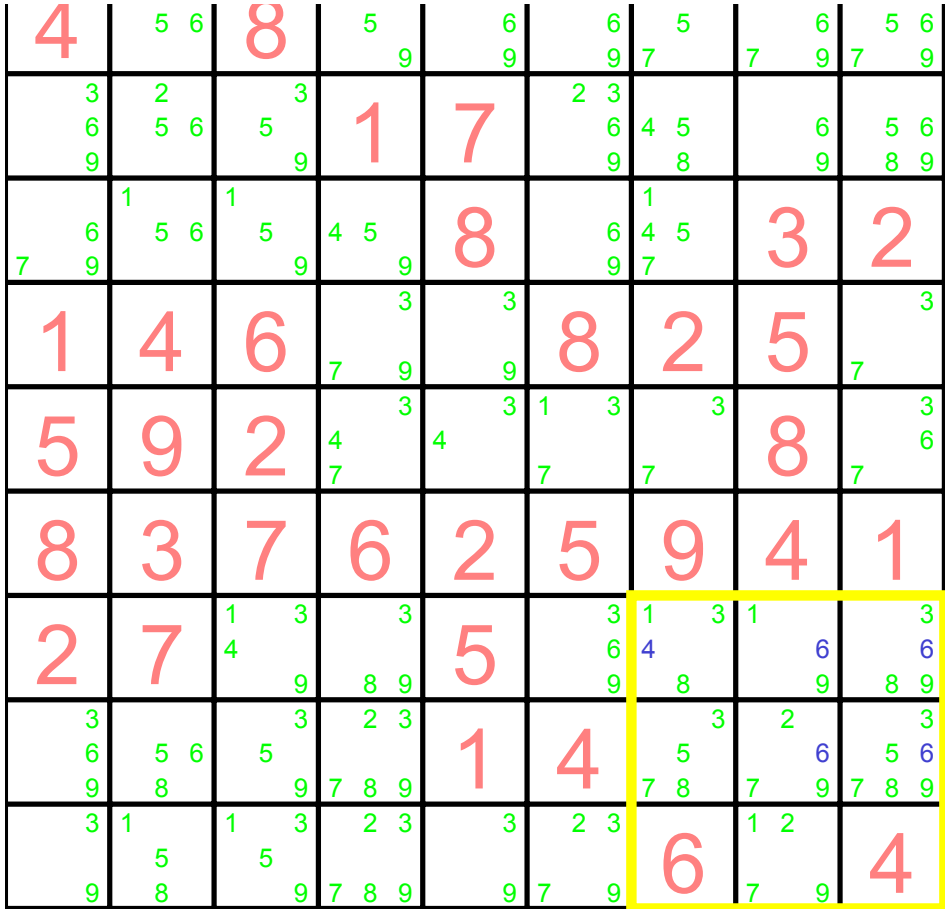
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4	5 6	8	5 9	6 9	6 9	5 7	6 9 7	5 6 7 9
3 6 9	2 5 6	3 5 9	1	7	2 3 6 9	4 5 8	6 9	5 6 8 9
6 7 9	1 5 6	1 5 9	4 5 9	8	6 9	1 4 5 7	3	2
1	4	6	3 7 9	3 9	8	2	5	3 7
5	9	2	3 4 7	3 4	1 3 7	3 7	8	3 6 7
8	3	7	6	2	5	9	4	1
2	7	1 3 4 9	3 8 9	5	3 6 9	1 3 8	1 9	3 8 9
3 6 9	5 6 8	3 5 9	2 3 7 8 9	1	4	3 5 7 8	2 7 9	3 5 7 8 9
3 9	1 5 8	1 3 5 9	2 3 7 8 9	3 9	2 3 7 9	6	1 2 7 9	4

Back to Start

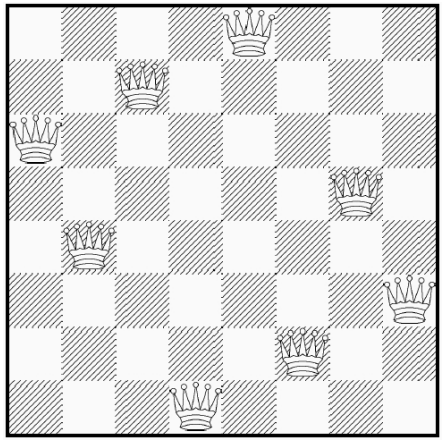
Sudoku

4		8						
			1	7				
				8			3	2
		6			8	2	5	
	9						8	
	3	7	6			9		
2	7			5				
					1	4		
						6		4

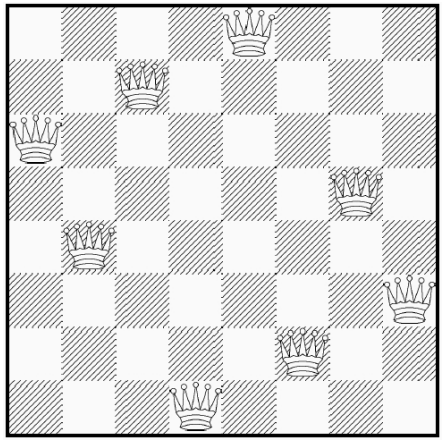
Solution

4	2	8	5	6	3	1	7	9
3	5	9	1	7	2	4	6	8
7	6	1	4	8	9	5	3	2
1	4	6	3	9	8	2	5	7
5	9	2	7	4	1	3	8	6
8	3	7	6	2	5	9	4	1
2	7	4	9	5	6	8	1	3
6	8	3	2	1	4	7	9	5
9	1	5	8	3	7	6	2	4

each other.

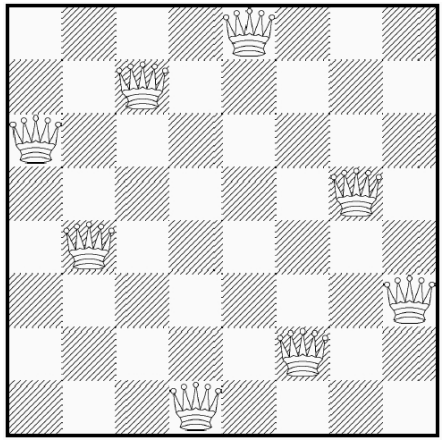


each other.



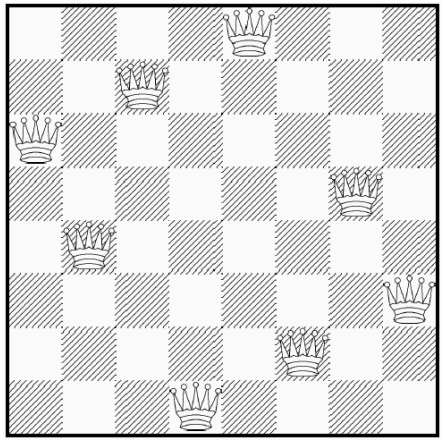
- Variables: $X_1, \ldots, X_N$ each variable for each column
- Domains: domain($X_1, \ldots, X_N$],1,N)

each other.



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- Domains: domain($X_1, \ldots, X_N$],1,N)
- Constraints: for $i \in [1..(N-1)]$ and $j \in [(i+1)..N]$:
  - $X_i \ \#\backslash= X_j$ (rows) $\Rightarrow$ all_different $([X_1, \ldots, X_N])$

each other.



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  - $X_i \ \#\backslash= X_j$ (rows) $\Rightarrow$ all_different $([X_1, \ldots, X_N])$
  - $X_i - X_j \ \#\backslash= i - j$ (diagonal)
    e.g. X_1=1 and X_2=2 prohibited ... diagonal through (1,1) a (2,2)
  - $X_i - X_j \ \#\backslash= j - i$ (diagonal)
    e.g. X_1=3 and X_2=2 prohibited ... diagonal through (1,3) a (2,2)

Applications



Applications

- Operation research (optimization problems)
- Graphical interactive systems (to express geometrical correctness)
- Molecular biology (DNA sequencing, 3D models of proteins)
- Finance
- Circuit verification
- Elaboration of natural languages (construction of efficient parsers)
- Scheduling of activities
- Configuration problem in form compilation
- Generation of coerent music programs **?**.
- Data bases
- ...
- http://hsimonis.wordpress.com/

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
Period 1	0 vs 1	0 vs 2	4 vs 7	3 vs 6	3 vs 7	1 vs 5	2 vs 4
Period 2	2 vs 3	1 vs 7	0 vs 3	5 vs 7	1 vs 4	0 vs 6	5 vs 6
Period 3	4 vs 5	3 vs 5	1 vs 6	0 vs 4	2 vs 6	2 vs 7	0 vs 7
Period 4	6 vs 7	4 vs 6	2 vs 5	1 vs 2	0 vs 5	3 vs 4	1 vs 3

Schedule of 1997/1998 ACC basketball league (9 teams)

- various complicated side constraints
- all 179 solutions were found in 24h using enumeration and integer linear programming [Nemhauser & Trick, 1998]
- all 179 solutions were found in less than a minute using constraint programming [Henz, 1999, 2001]



- Gate allocation at the Hong Kong International Airport
- System was implemented in only four months, and includes constraint programming technology (ILOG)
- Schedules ~900 flights a day (53 million passengers in 2011)





- One of the world's largest container transshipment hubs
- Links shippers to a network of 200 shipping lines with connections to 600 ports in 123 countries
- Problem: Assign yard locations and loading plans under various operational and safety requirements
- Solution: Yard planning system, based on constraint programming (ILOG)

- Netherlands Railways has among the densest rail networks in the world, with 5,500 trains per day
- Constraint programming is one of the components in their railway planning software, which was used to design a new timetable from scratch (2009)
- Much more robust and effective schedule, and $75M additional annual profit
- INFORMS Edelman Award winner (2009)

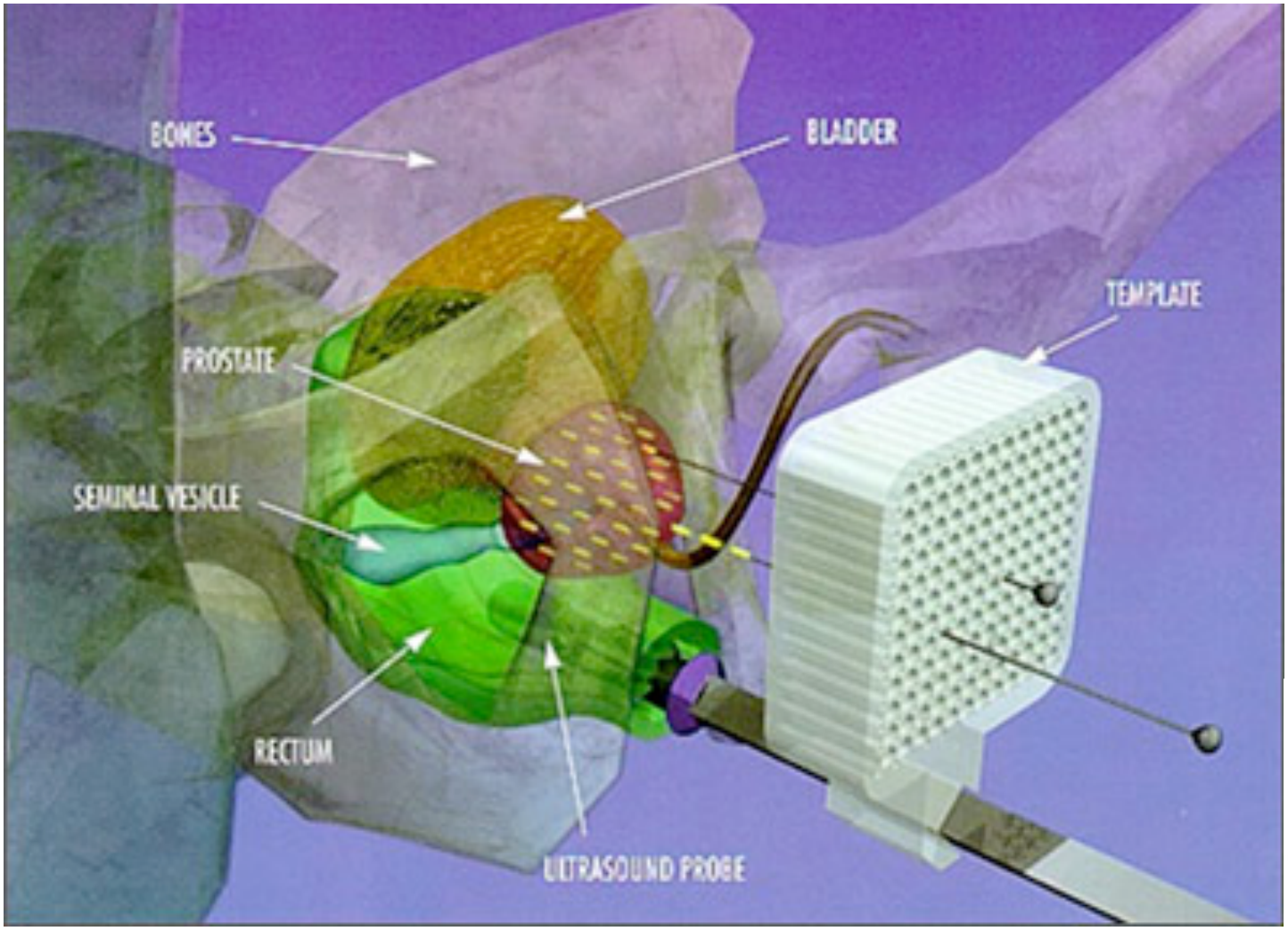


What is it?

- Radiation therapy uses ionising radiation in the treatment of patients diagnosed with cancer.
- The aim of radiation therapy is to deliver a precisely measured dose of radiation to a well-defined tumor volume whilst sparing the surrounding normal tissue.

Variants for which optimisation is key

- **IMRT** we've discussed this earlier.
- **Brachytherapy** involves the placement of radioactive sources directly into tissue.


BONES
BLADDER
TEMPLATE
PROSTATE
SEMINAL VESICLE
RECTUM
ULTRASOUND PROBE



- The input is a 3-D dosage matrix of lower bounds and upper bounds on radiation exposure.
- Boolean variables encode the (non-)placement of seeds in a 3-D grid of potential locations.
- The treatment plan comprises an assignment to the variables of to represent the (non)placement of seeds in the three-dimensional grid of potential locations.

Conclusion



Conclusion

- AI Applications: modeling and solving:
  - Resource Allocation (Scheduling).
  - Planning.
  - Biology (DNA sequencing).
  - Data Mining and Machine Learning.
- Advantages:
  - Easy problem modeling
  - Flexible treatment of variants of original problems.
  - Easy control of the search
- Limitations:
  - Over-Constrained problems.
  - Interactivity.


Beauty is born
from constraints
Leonardo Da Vinci